Conditional LoRA Bridges: Separating Adaptation Capacity from Sensor Conditioning

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September 07, 2026 — revision 2

Abstract

A hypernetwork can condition a frozen language model by generating low-rank weight updates from sensor features. The same bridge also adds trainable capacity, so improved text likelihood alone is an ambiguous indicator of sensor use. We compare separate training cohorts for unpaired text, paired category labels, and sensor-plus-instruction responses. We use the original 33.6M host and fixed versions of 230M and 0.8B hosts, with matched constant, shuffled, random, static-adapter, and frozen references. Under independent sampling, 9 of 9 true-minus-constant BPB intervals include zero. This does not prove equivalence; paired likelihood objectives may still benefit from conditioning. We assess conditioning on full held-out populations, using grouped intervals and sensor interventions on the same checkpoints. Numerical training failures affected 25 of the 411 declared trials; the corresponding endpoint estimates are unavailable. Mini reaches macro rank-1 accuracy 0.461 on audio, 0.582 on vision; Qwen 0.8B reaches macro rank-1 accuracy 0.838 on IMU, 0.421 on audio, 0.451 on vision. On held-out templates, Qwen's IMU joint label and sentence success is 0.838 and 0.689; sensor replacement tests their input dependence. Success on both reversed rule mappings is 0.005. Audio generation varies substantially across seeds. We measure changes over training, variation in effective updates, three composition operators, and bridge costs. The dense output head grows with adapted dimensions. Workstation measurements leave edge efficiency untested. The original audio encoder used held-out inputs for normalization, which limits the audio evidence. The complete evidence and reproduction code accompany the paper.

1 Introduction

A frozen language model can receive sensor information through changes to its effective weights. A hypernetwork can map an external encoder's features to low-rank updates, while both the encoder and the language model remain frozen. This modular interface also adds trainable parameters. Better text likelihood can therefore reflect ordinary text adaptation without using changes in the sensor input.

We study this distinction with three separately trained cohorts. Cohort U uses independently sampled text and sensor features. Cohort P uses paired sensor and category examples. Cohort G uses paired sensors, instructions, and target responses. U tests text adaptation under a deliberately unpaired objective; P tests closed-set label ranking; G tests sensor-dependent labels, sentences, and binary rule responses. The cohorts share an interface and controls but use different supervision and weights. We also measure each saved checkpoint under several input interventions. Results from a task-trained bridge therefore remain separate from those of a bridge trained only on unpaired text.

The interpretation depends on how sensors and target text are sampled. With paired data, conditioning can reduce text cross-entropy. U samples text and sensors independently, so shuffling cannot disrupt designed semantic pairs. A constant-feature bridge tests whether variable sensor input is necessary for the observed improvement in U. A small or statistically uncertain difference between true and constant features does not prove equivalence.

We extend the original mini-model study to two larger hosts with fixed model versions. We use full held-out task populations, uncertainty across seeds and clusters, vision probes, generation tests, effective-update diagnostics, and resource measurements. These comparisons test when the bridge uses sensor information and how its behavior and cost limit its use. The tasks cover label ranking and constrained responses; they do not benchmark a general multimodal assistant.

2 Interface and controls

Conditional low-rank updates. Let $f = e(s)$ be the feature vector of a frozen sensor encoder. The bridge h_θ produces $w(f) \in \mathbb{R}^D$, partitioned into factors $A_i(f) \in \mathbb{R}^{d_i^{\text{in}} \times r}$ and $B_i(f) \in \mathbb{R}^{r \times d_i^{\text{out}}}$ for each target module. The module computes

$$y = xW_i + \frac{\alpha}{r} xA_i(f)B_i(f), \qquad D = r \sum_i (d_i^{\text{in}} + d_i^{\text{out}}).$$

Each batch element receives its own factors. Candidate labels for one sensor share only that sensor's factors. We train the bridge through the frozen host; the host and external encoder parameters receive no updates.

The dense bridge projects the input to 64 dimensions, concatenates a trainable 64-dimensional vector shared across inputs, applies two width-256 hidden layers with GELU, and emits the flattened factors. We use rank $r = 4$ and scale $\alpha = 4$. Its last layer has $257D$ parameters, including its bias. Thus its absolute size grows with the adapted dimensions and target count. Its ratio to the host size need not grow. Table 1 gives the parameter counts of the instantiated models.

Controls. The true, constant, shuffled, and random conditions use the same bridge architecture. The constant is the mean of the relevant training feature pool. It provides an architecture- and parameter-matched training control. Its input also stays constant in U's BPB evaluation. The shuffled paired-training condition rotates each sampled batch; the random condition replaces features with independent Gaussian draws. All trained controls receive correct features for the primary ranking test. Separate interventions on the same true-trained checkpoint replace its test features with a constant, random features, a sensor from a different class, five fixed permutations, or no bridge. These interventions test the effect of changing inputs after training. Presenting correct features to a constant- or random-trained bridge can also change its input distribution. These comparisons test the training signal; they do not isolate the effect of corrupting test inputs. We interpret them alongside the static adapter and interventions on the true-trained checkpoint.

Static LoRA directly optimizes factors on the identical host target map. It receives no sensor input and has fewer trainable parameters than the dense bridge. We also test eight sensor-generated *input-prefix embeddings*. These are concatenated before the text embeddings, as an input-level soft prompt related to prompt tuning [8]. Li and Liang [9] instead parameterize prefixes at each layer. Both the prefix and its reference score the same 128 text positions. We report the initialized prefix and its trained endpoints. The results apply to the tested parameterization, length, optimizer, and budget.

Diversity and functional dependence. For a batch of N generated factor vectors, training can add

$$\mathcal{L}_{\text{div}} = \frac{1}{N(N-1)} \sum_{j \neq k} \cos(w(f_j), w(f_k)), \qquad \mathcal{L} = \mathcal{L}_{\text{text}} + \lambda \mathcal{L}_{\text{div}}.$$

The registered IMU sweep is $\lambda \in \{0, .01, .05, .10, .20\}$ for U and P. Factor cosine alone is insufficient: equivalent low-rank updates can have different factorizations. We also measure the geometry of the effective updates $A_i B_i$ and the KL divergence of next-token distributions from their mean over sensors, with the text held fixed. Neither geometric variation nor a nonzero KL value alone establishes correct sensor behavior.

3 Experimental protocol

Hosts. We use the original 33.6M-parameter mini checkpoint, LFM2.5-230M [10], and the language tower of Qwen3.5-0.8B [17]. LFM and Qwen are post-trained released models. The mini uses an 8192-token vocabulary; the larger hosts retain their native tokenizers and special-token policies. LFM targets include attention projections and feed-forward layers. Qwen targets include full-attention projections and feed-forward layers; its DeltaNet projections and native vision tower are not adapted. No native image tokens enter the vision probe. The feature bridge is its only image-dependent input path.

Table 1: Parameter counts for the instantiated language models and IMU bridges. M denotes million parameters. Audio and vision use different small input projections; the resource artifact gives their exact counts. These counts exclude the native Qwen vision tower.

Host	Host M	D	Targets	Bridge M	Bridge/host
Mini	33.56	327,680	56	84.33	2.51
LFM 230M	229.69	774,144	66	199.07	0.87
Qwen 0.8B	752.39	1,597,440	96	410.66	0.55

The hosts differ in architecture, tokenizer, pretraining, and post-training as well as parameter count, so this comparison cannot isolate a scaling law. We use one recipe without tuning it for each model, preserving rank, bridge shape, feature encoders, control definitions, learning rate, optimizer, batch sizes, and step budgets.

Frozen features and held-out units. IMU uses UCI-HAR's 7352 training windows and all 2947 test windows across nine held-out subjects [2]. Audio uses ESC-50 folds 1–4 for bridge fitting and all 400 fold-5 clips for testing [16]. Audio items retain their original-source identifiers. Vision uses the original frozen ImageNet MobileNetV3-Small and Big Buck Bunny perceiver with its original BGR preprocessing. The paired vision task uses CIFAR-10 [7]: 45000 images for bridge fitting, a fixed 5000-image development subset, and all 10000 test images. We do not use the development subset to select models or settings. The unpaired vision pool remains the original video frames.

Audio preprocessing limits the interpretation of the test results. The stored normalization matches statistics computed over all 2000 clips, including test clips. Two original sources also cross the supplied fold boundary. We keep the original encoder for this replication and report a secondary test analysis that excludes those sources. Their exclusion leaves the earlier normalization exposure in place. Complete records of the original encoder training run are unavailable. IMU normalization

matches its training split and its test subjects are disjoint. The image audit finds no exact RGB-array train/test duplicates, but does not exclude near-duplicates or unknown upstream pretraining exposure. Here, "held out" means excluded from bridge fitting, subject to these known encoder exposures.

Objectives and budgets. U trains on ClimbMix text shards 0, 1, and 2 with independently sampled features from fixed 64-item pools. It uses batch 8, 128 predicted text positions, and checkpoints at 300 and 3000 optimizer steps. The paired label cohort P uses batch 4 and the original training loss averaged over the whole prompt and label, with checkpoints at 600 and 6000 steps. G uses separate examples in the host's native chat format and computes loss only on the assistant response, including native EOS, at the same paired step budgets and batch size. Its examples must fit within 256 tokens; longer examples trigger an error.

All optimizers are AdamW with learning rate .001, $(\beta_1, \beta_2) = (.9, .999)$, $\epsilon = 10^{-8}$, weight decay .01, and gradient norm clipping at 1. The default seeds are 42, 1042, and 2042. Mini and Qwen IMU/audio P controls add 3042 and 4042. Vision P and generation G use three seeds. The IMU penalty sweep uses the three common seeds; paired comparisons with its zero-penalty reference use that same seed subset. Fixed checkpoints share a trajectory; they are not independent training replicates. Optimizer steps, examples, supervised tokens, and supervised bytes are retained in the artifact. Equal steps do not imply equal tokenization or compute across hosts.

The registered matrix contains 411 trials: 406 training trajectories and five frozen references with zero optimizer steps. The planned training budget is 1,896,000 optimizer updates; numerical terminations can shorten trajectories.

Execution uses FP32 parameters, optimizer state, and arithmetic with autocast and TF32 disabled. A BF16 LFM acceptance run failed the strict batch/scalar loss check; the FP32 path passed without widening its tolerance. Tests cover per-example factor isolation, gradient flow through a frozen host, one next-token shift, natural prompt/label boundaries, byte accounting, cache reset after sensor changes, and optimizer/RNG resume. These checks apply to the reported hardware and locked software environment. The same seed may produce different results on another backend.

We stop a training run if the loss or gradient norm is nonfinite before an optimizer update. A nonfinite norm can occur even when individual gradient elements are finite. Numerical failure also differs from loss of conditioning, which requires finite checkpoint evaluations of task behavior and sensor dependence. We evaluate checkpoints saved before failure and mark later endpoints unavailable. An aggregate requires all its declared seeds; we do not substitute earlier weights or average only successful seeds. We added this failure handling after the first numerical failure, following the preparation plan's reporting rule and preserving the registered training operations. Hardware and infrastructure errors are recorded separately.

Likelihood and ranking. Native BPB is total negative log-likelihood divided by $\log 2$ times total target bytes. Special tokens contribute no bytes. The short and long panels contain 256 and 1024 packed validation rows from shard 6542. These panels are paired within a host. A separate 512-document common-byte panel scores identical raw-text target spans and shared prefix bytes across tokenizers. We keep native packing and common-byte comparisons separate.

P ranking compares the mean log-likelihood of candidate label tokens after the original modality prompt. Tokenization can merge across that boundary; the score includes the boundary-crossing label token exactly once. Paired-label BPB counts that whole token's bytes, including any prompt-boundary bytes it contains; it is a within-host diagnostic, separate from the common-byte panel.

Ties follow the fixed category order. We report macro and micro rank-1 accuracy, top-2, average rank, and paired-label BPB. The original 64-item procedure remains a secondary size diagnostic. Three fixed prompt paraphrases test robustness without further training. Uniform random-choice accuracy is $1/K$ for K classes; an actual frozen model's micro accuracy need not equal that reference on an imbalanced test set.

We evaluate intermediate IMU checkpoints on the full ranking test population and use fixed subsets of test features for geometry. These trajectories are descriptive; the short and long endpoints remain the primary comparisons. We include every penalty and saved point without selecting a test-set peak. The artifact also gives the correct-label score minus the strongest alternative and the entropy of the softmax-normalized candidate scores. These scores are mean label log-likelihoods, not calibrated class probabilities.

Generation. First, we decode Qwen P checkpoints at $\lambda = 0$ without further training, using the original ranking prompt. We then evaluate G checkpoints on category-only responses, short sentences, and explicit binary rules. Each family has three training and three held-out prompt templates. Evaluation includes one training template and all held-out templates. Rule queries match the sensor category in half the cases; each mapping is also reversed with the same sensor. This tests whether the response can depend on both the sensor and the supplied rule.

The generation panel uses all 400 audio clips and 300 IMU windows chosen by class and subject before inspecting model outputs. The P vision decoding diagnostic uses 300 images. G has 16 instruction cases per sensor; these are 16 repeated measurements of the same sensor. All decoding is greedy and limited to 64 new tokens. G stops at the pinned model EOS (`248044`) or assistant-end token (`248046`). Raw P decoding retains the host's original stopping policy. We report semantic correctness, format compliance, and joint success separately, and retain every response. For category and sentence tasks, the semantic score is a deterministic longest-category-phrase match after case and underscore normalization; it can credit a category mention in an otherwise incorrect sentence. Joint success requires the exact normalized target and is the primary response score. Rule correctness requires the exact mapped word. We also report a secondary rule score that requires both reversed mappings for the same sensor and template to be correct. Repeated templates are then averaged within a sensor before uncertainty estimation. We also prompt the frozen host with either the oracle label or the feature classifier's label. These references supply label text in addition to the ordinary prompt.

Uncertainty. Reported 95% intervals use 10000 crossed seed/cluster bootstrap replicates, with equal weight per training seed. Clusters are IMU subjects, original audio sources, and images. Resampling is stratified by class when each cluster belongs to one class; subjects spanning activities are sampled jointly. Paired contrasts preserve the same seeds and held-out units. Likelihood uses byte-weighted text-unit resampling. Generation first groups repeated prompts by sensor. Intervals are descriptive; we make no significance claims across multiple comparisons and specify no equivalence margin. The many IMU windows come from only nine test subjects, which limits population coverage.

4 Results

25 of the 411 declared trials terminated with nonfinite loss or gradient norms before an optimizer update (Table 13). We keep these failures as outcomes of the specified training settings. We do not

substitute earlier checkpoints or retry seeds until they succeed. A dash marks an estimate missing one or more declared seeds. The evidence includes every available per-seed measurement.

4.1 Unpaired text adaptation and input prefixes

Table 2 reports unpaired adaptation at 3000 steps. Across 9 host/modality pairs, the true-minus-constant BPB differences range from -0.0010 to 0.0005; 9 of their 9 intervals include zero. These comparisons test whether variable features are needed when text and sensors are sampled independently within each host. Intervals spanning zero do not prove equivalence. Paired sensor/text data could still yield a likelihood benefit from conditioning. Static adapters measure text adaptation without sensor input. For true-minus-static BPB, 5 intervals are above zero, 2 below zero, and 2 include zero. Negative differences favor the bridge, whose performance relative to static adapters varies across these pairs. Figure 1 includes every training corruption at both budgets; the machine-readable summary retains the paired intervals, and the appendix reports the separate common-byte panel.

Table 2: Unpaired native BPB at 3000 steps (1024 shared rows per host). Lower is better. True-feature intervals include seeds and text units; complete control intervals and both budgets are in the appendix.

Host	Task	Frozen	True [95% interval]	Const.	Static
Mini	IMU	1.954	1.943 [1.925, 1.961]	1.943	1.941
Mini	Audio	1.954	1.942 [1.924, 1.960]	1.942	1.941
Mini	Vision	1.954	1.942 [1.924, 1.960]	1.942	1.941
LFM 230M	IMU	1.145	1.068 [1.052, 1.084]	1.067	1.066
LFM 230M	Audio	1.145	1.067 [1.051, 1.083]	1.067	1.066
LFM 230M	Vision	1.145	1.068 [1.052, 1.084]	1.067	1.066
Qwen 0.8B	IMU	0.998	0.986 [0.972, 1.001]	0.986	0.987
Qwen 0.8B	Audio	0.998	0.985 [0.971, 0.999]	0.985	0.987
Qwen 0.8B	Vision	0.998	0.984 [0.970, 0.998]	0.985	0.987

Table 3: Eight generated input-prefix embeddings: native BPB at initialization and both budgets. Initialization and 300 steps use 256 rows; 3000 steps use 1024. Compare each score with the frozen reference on the same panel; the panels differ across budgets. Each score includes the same 128 text positions as its reference.

Host	Task	Initialized	300 steps	3000 steps
Mini	IMU	1.965 [1.928, 2.002]	1.965 [1.928, 2.001]	1.953 [1.934, 1.971]
Mini	Audio	1.965 [1.928, 2.002]	1.964 [1.927, 2.001]	1.953 [1.934, 1.971]
Mini	Vision	1.965 [1.929, 2.002]	1.964 [1.927, 2.001]	1.953 [1.934, 1.971]
LFM 230M	IMU	1.192 [1.158, 1.226]	1.114 [1.082, 1.146]	1.070 [1.055, 1.086]
LFM 230M	Audio	1.198 [1.163, 1.232]	1.114 [1.082, 1.146]	1.072 [1.057, 1.088]
LFM 230M	Vision	1.192 [1.158, 1.225]	1.115 [1.083, 1.147]	1.074 [1.058, 1.090]
Qwen 0.8B	IMU	1.299 [1.253, 1.344]	1.004 [0.975, 1.033]	0.962 [0.948, 0.976]
Qwen 0.8B	Audio	1.280 [1.219, 1.347]	1.004 [0.975, 1.033]	0.963 [0.949, 0.977]
Qwen 0.8B	Vision	1.358 [1.303, 1.414]	1.004 [0.976, 1.033]	0.964 [0.949, 0.978]

At 3000 U steps, prefix-minus-frozen BPB differences range from -0.0742 to -0.0014; 9 of 9 paired intervals are below zero. Table 3 also shows initialization and the short budget. Prefix and

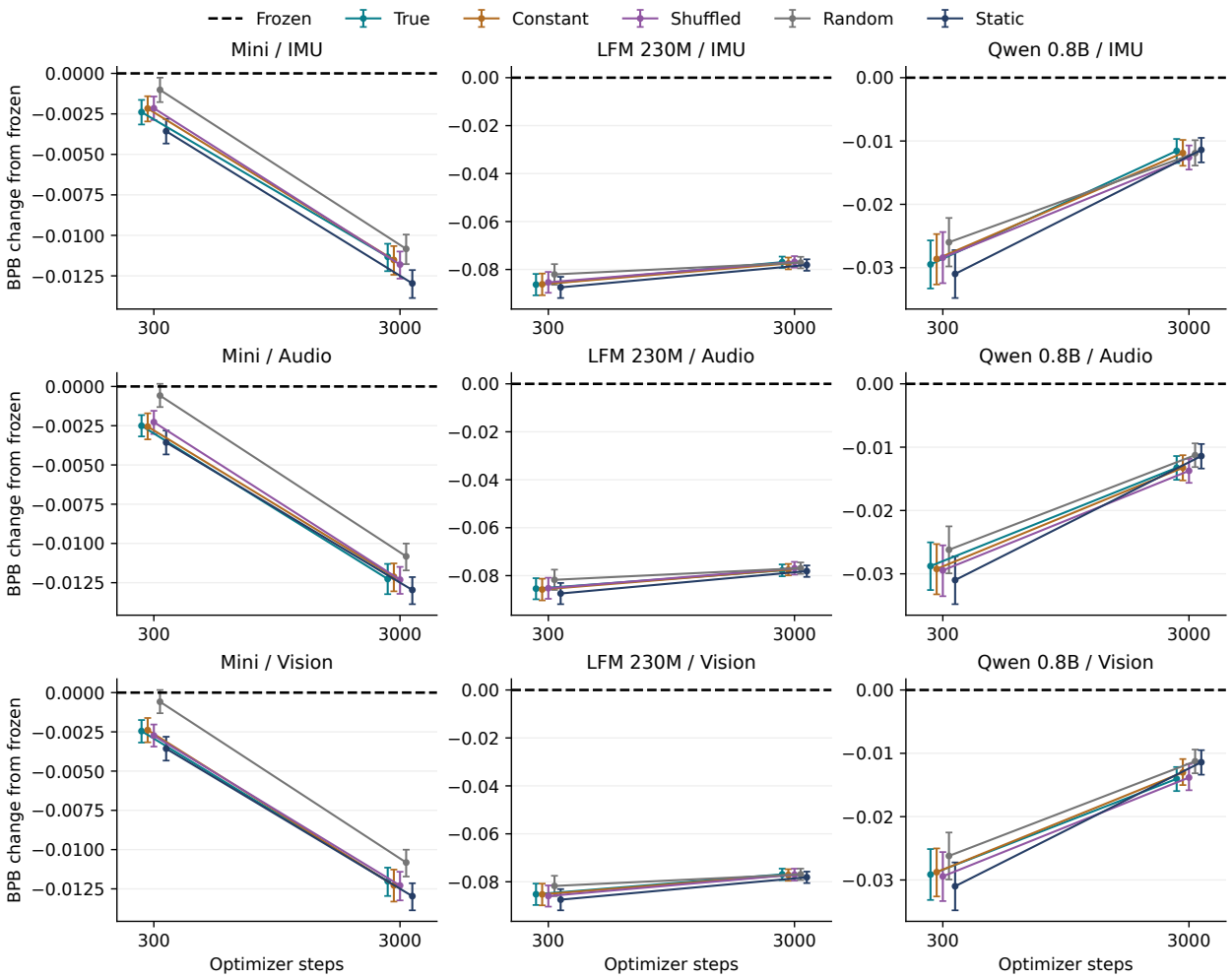


Figure 1: Unpaired text adaptation at the two fixed budgets. Values are paired native-BPB differences from the frozen host on the matching evaluation panel; negative values improve likelihood. The 300-step panel has 256 rows and the 3000-step panel has 1024, so the changes at each budget use different test samples. All plotted conditions have their registered seeds. The common-byte table uses a separate fixed panel at both budgets.

reference score the same 128 text positions. The result covers eight input-prefix embeddings trained on U. It does not show sensor use, and the separately trained P and G bridges provide no matched comparison of prefix conditioning.

4.2 Held-out sensor ranking

Table 4 uses all held-out items, with seed and cluster intervals. The frozen-feature linear classifier reaches macro accuracy IMU 0.967 [0.934, 0.989]; Audio 0.662 [0.624, 0.700]; Vision 0.799 [0.791, 0.807]. This reference measures label information already available in the encoder features, against which to compare the bridge. We also replace the correct features with the first fixed permutation while holding each true-trained checkpoint fixed. The macro-accuracy differences (correct minus permuted) are Mini/Audio 0.443 [0.403, 0.486]; Mini/Vision 0.481 [0.426, 0.527]; LFM 230M/IMU 0.686 [0.611, 0.758]; LFM 230M/Audio 0.368 [0.317, 0.424]; Qwen 0.8B/IMU 0.685 [0.582, 0.775];

Qwen 0.8B/Audio 0.402 [0.363, 0.438]; Qwen 0.8B/Vision 0.351 [0.248, 0.455]. The evidence includes all five fixed permutations and the other feature interventions. We show the first permutation without selecting it by its result.

The new vision probes give Mini rank-1 accuracy 0.582 [0.526, 0.628], correct-minus-different-class difference 0.536 [0.474, 0.586]; Qwen 0.8B rank-1 accuracy 0.451 [0.353, 0.553], correct-minus-different-class difference 0.389 [0.280, 0.505]. These interventions test whether image features affect ten-class label ranking, which U's BPB result cannot determine. They do not test visual generation or open-world reasoning.

Table 4: Full-population macro rank-1 accuracy at 6000 P steps, with no diversity penalty. Uniform-choice references are 1/6 (IMU), 1/50 (audio), and 1/10 (vision). Mini/Qwen IMU/audio use five seeds; other paired cells use three. LFM vision P is outside the registered grid.

Host	Task	Frozen	Static	Const.	Shuf.	Rand.	True [95%]
Mini	IMU	0.167	0.167	–	–	0.190	–
Mini	Audio	0.020	0.020	0.020	0.020	0.021	0.461 [0.419, 0.506]
Mini	Vision	0.100	0.100	0.100	–	0.112	0.582 [0.526, 0.628]
LFM 230M	IMU	0.167	0.167	0.167	0.176	0.109	0.839 [0.761, 0.913]
LFM 230M	Audio	0.020	0.020	0.018	0.020	0.023	0.386 [0.336, 0.442]
Qwen 0.8B	IMU	0.167	0.167	0.167	0.173	0.175	0.838 [0.733, 0.927]
Qwen 0.8B	Audio	0.020	0.020	0.021	0.020	0.023	0.421 [0.383, 0.459]
Qwen 0.8B	Vision	0.100	0.100	0.100	0.100	0.100	0.451 [0.353, 0.553]

Table 5: Test interventions on identical true-trained P checkpoints at 6000 steps and zero penalty. Values are macro accuracy. Different-class substitution guarantees a changed class; a permutation can retain it. Paired intervals, random features, and all five permutations are in the machine-readable analysis.

Host	Task	Correct	Constant	Off	Other class	Shuffle 0
Mini	IMU	–	–	–	–	–
Mini	Audio	0.461	0.020	0.020	0.010	0.019
Mini	Vision	0.582	0.100	0.100	0.047	0.101
LFM 230M	IMU	0.839	0.167	0.167	0.032	0.152
LFM 230M	Audio	0.386	0.020	0.020	0.011	0.018
Qwen 0.8B	IMU	0.838	0.167	0.167	0.032	0.153
Qwen 0.8B	Audio	0.421	0.020	0.020	0.013	0.020
Qwen 0.8B	Vision	0.451	0.100	0.100	0.062	0.100

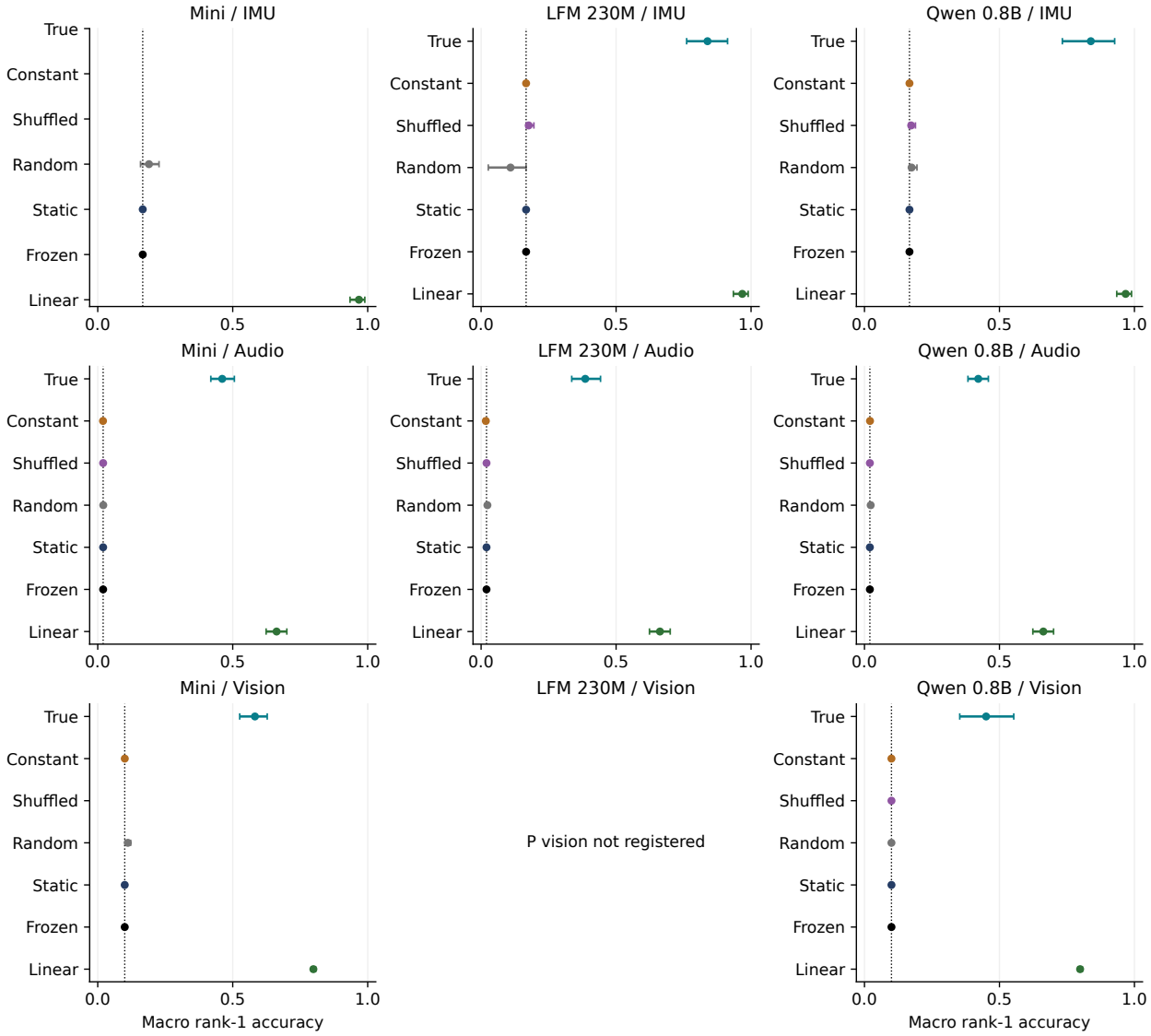


Figure 2: Macro rank-1 accuracy for the adapters trained on paired P examples at 6000 steps. The full test population and grouped intervals replace the original 64-item primary evaluation. Frozen feature classifiers measure information already available in the encoder features.

4.3 Budget, regularization, and conditioning

Figure 3 connects each IMU P accuracy curve to effective-update cosine and fixed-text output sensitivity from the same weights. Table 6 compares 600 with 6000 steps using only seeds 42, 1042, and 2042 at every penalty.

Table 6: IMU P macro rank-1 accuracy and grouped 95% intervals for the three shared seeds. The zero-penalty row uses the same seed subset as every other penalty; additional primary probe seeds do not enter these curves. Dashes mark unavailable full-seed estimates.

Host	λ	600 steps [95%]	6000 steps [95%]
Mini	0	0.741 [0.693, 0.784]	–
Mini	0.01	0.802 [0.770, 0.830]	–
Mini	0.05	–	–
Mini	0.1	0.759 [0.701, 0.804]	–
Mini	0.2	0.804 [0.725, 0.903]	–
LFM 230M	0	0.834 [0.759, 0.911]	0.839 [0.761, 0.913]
LFM 230M	0.01	0.723 [0.644, 0.809]	0.768 [0.645, 0.869]
LFM 230M	0.05	0.759 [0.651, 0.908]	0.741 [0.652, 0.802]
LFM 230M	0.1	0.709 [0.623, 0.794]	0.676 [0.640, 0.729]
LFM 230M	0.2	0.733 [0.628, 0.820]	0.737 [0.658, 0.840]
Qwen 0.8B	0	0.862 [0.747, 0.956]	0.788 [0.655, 0.911]
Qwen 0.8B	0.01	0.738 [0.638, 0.833]	0.607 [0.553, 0.649]
Qwen 0.8B	0.05	0.756 [0.690, 0.821]	0.762 [0.653, 0.867]
Qwen 0.8B	0.1	0.667 [0.484, 0.820]	0.762 [0.692, 0.834]
Qwen 0.8B	0.2	0.820 [0.680, 0.923]	0.838 [0.718, 0.929]

Numerical failures leave the three-seed 6000-step estimates incomplete for Mini (5 of 5 penalty settings). Earlier per-seed measurements remain available. At zero penalty, the 600-to-6000-step comparison is LFM 230M 0.834 [0.759, 0.911] to 0.839 [0.761, 0.913]; Qwen 0.8B 0.862 [0.747, 0.956] to 0.788 [0.655, 0.911]. The available intervals at the long endpoint for larger hosts lie above six-class chance accuracy. Collapse to chance is therefore not universal at this budget.

At the largest declared penalty, $\lambda = 0.2$, and 6000 P steps, the seed means are LFM 230M: factor cosine 0.145, effective-update cosine 0.694, and output KL 0.979 nats; Qwen 0.8B: factor cosine 0.009, effective-update cosine 0.790, and output KL 1.396 nats. The three measures describe variation in factors, effective-update directions, and outputs, respectively. Cosine omits update magnitude; the full analysis also gives centered energies and norms. Penalty effects differ across hosts and budgets. The full U trajectories are retained in the analysis. We do not select a penalty by its test-set peak. Results at the short budget cannot define a collapse threshold that applies generally.

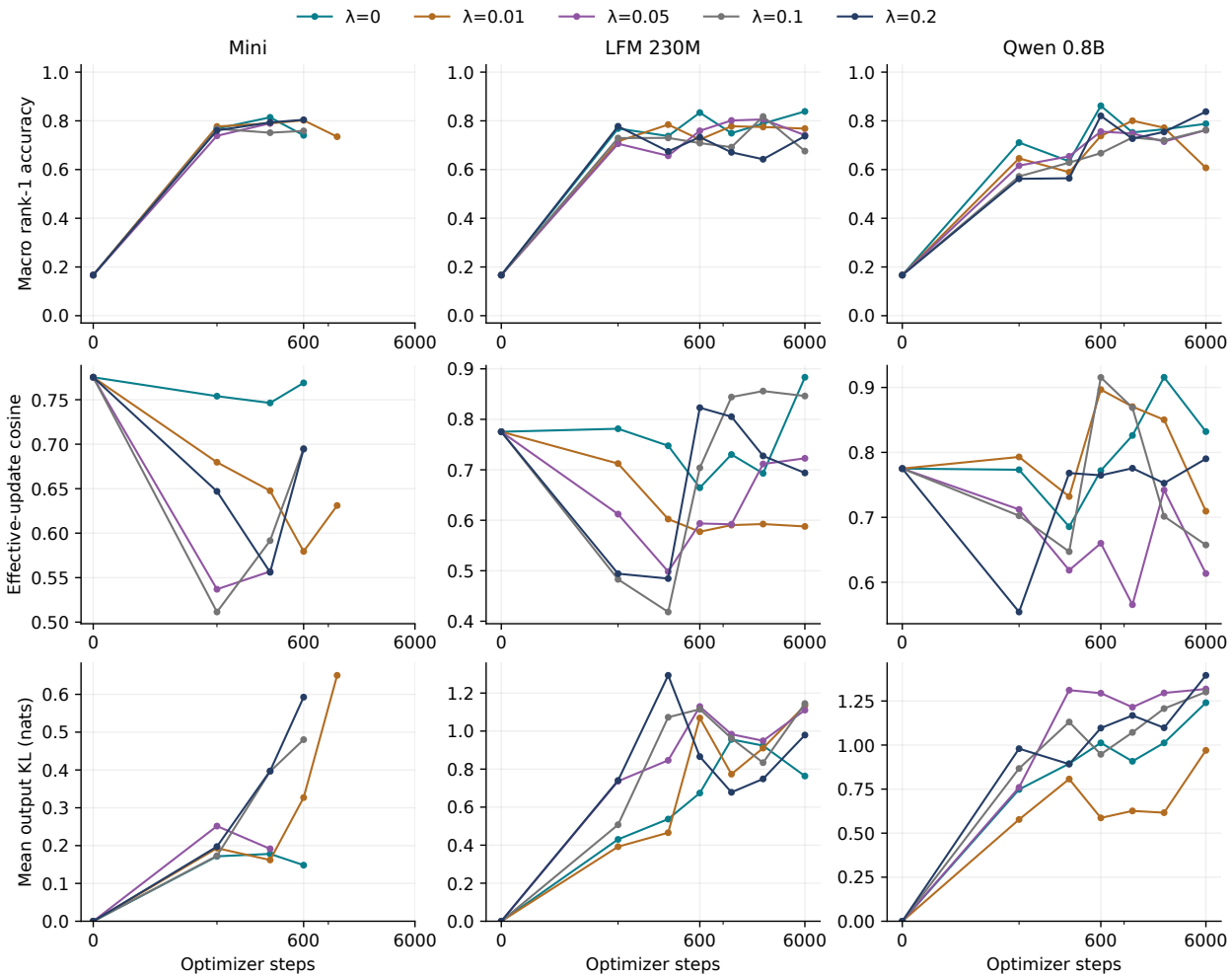


Figure 3: IMU P trajectories with the registered diversity penalties. Accuracy, effective-update geometry, and output sensitivity use the same saved weights at each point. Intermediate checkpoints are repeated measurements of a trajectory, not additional seeds. All penalties use three shared seeds. Only complete three-seed estimates are drawn; curves end when numerical stops prevent later estimates. Endpoint intervals are in Table 6.

4.4 From ranking to generated responses

Tables 7–9 distinguish raw P decoding, separately trained instruction responses, and label-in-text references. All values below are class-macro scores on held-out templates at 6000 G steps.

For IMU, label success is 0.838 [0.737, 0.936] (correct-minus-different-class 0.832 [0.731, 0.933]); sentence success is 0.689 [0.198, 0.971] (correct-minus-different-class 0.686 [0.198, 0.966]). Per-seed label scores are 0.928, 0.740, 0.847; sentence scores are 0.916, 0.193, 0.959, in seed order 42, 1042, 2042.

For Audio, label success is 0.056 [0.000, 0.161] (correct-minus-different-class 0.051 [0.000, 0.144]); sentence success is 0.064 [0.000, 0.186] (correct-minus-different-class 0.060 [0.000, 0.167]). Per-seed label scores are 0.000, 0.000, 0.169; sentence scores are 0.000, 0.000, 0.192, in seed order 42, 1042, 2042. Two of the three seeds have zero exact success in both families.

For held-out rule prompts, the differences in macro-averaged joint success (correct sensor minus

a sensor from another class) are IMU -2.1×10^{-4} [-0.007, 0.004]; Audio -0.001 [-0.007, 0.006]. Requiring both reversed mappings to be correct for the same sensor and template gives the secondary scores IMU 0.005 [0.001, 0.013]; Audio 0.001 [0.000, 0.002]. Reversing the output mapping while keeping the sensor fixed tests whether responses follow the supplied rule as well as the sensor. These deterministic templates cannot demonstrate general instruction following, and sensor-dependent labels or fixed sentences alone do not show reliable rule application. The label-in-text references are practical comparisons with a different input interface; their architectures are not matched to the bridge. The evidence includes every response, including invalid and repetitive output.

Table 7: Qwen G at 6000 steps: class-macro response scores on held-out instruction templates. Each estimate requires all three training seeds. Joint success requires the exact normalized target; semantic and format scores identify different failure types.

Task	Family	Semantic [95%]	Format [95%]	Joint [95%]
IMU	Label	0.877 [0.748, 0.979]	0.892 [0.799, 0.987]	0.838 [0.737, 0.936]
IMU	Sentence	0.866 [0.729, 0.971]	0.731 [0.196, 1.000]	0.689 [0.198, 0.971]
IMU	Rule	0.501 [0.497, 0.505]	0.999 [0.998, 1.000]	0.501 [0.497, 0.505]
Audio	Label	0.058 [0.000, 0.166]	0.224 [0.000, 0.668]	0.056 [0.000, 0.161]
Audio	Sentence	0.064 [0.000, 0.185]	0.281 [0.000, 0.838]	0.064 [0.000, 0.186]
Audio	Rule	0.158 [0.000, 0.468]	0.316 [0.000, 0.937]	0.158 [0.000, 0.468]

Table 8: Generation references: joint success averaged by class on held-out templates, using the same G prompts and decoding. Oracle and classifier prompts give the frozen host an explicit label; other prompts supply no label text. The full analysis includes intervals and both budgets.

Task	Family	Frozen	Const.	Shuf.	Rand.	Static	Oracle	Classifier
IMU	Label	0.000	0.167	0.167	0.167	0.148	0.778	0.757
IMU	Sentence	0.000	0.159	0.139	0.205	0.130	0.500	0.484
IMU	Rule	0.162	0.498	0.460	0.500	0.446	0.504	0.502
Audio	Label	0.000	0.016	0.020	0.022	0.020	0.587	0.381
Audio	Sentence	0.000	0.017	0.013	0.019	0.020	0.533	0.347
Audio	Rule	0.226	0.500	0.500	0.499	0.500	0.531	0.465

Table 9: Raw generation from Qwen P checkpoints at 6000 steps, without additional instruction training. The frozen reference uses the identical original ranking prompt. Exact category output is joint success, averaged by class. This diagnostic is separate from native-chat G training.

Task	Frozen [95%]	P static [95%]	P true bridge [95%]
IMU	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Audio	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]
Vision	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]	0.000 [0.000, 0.000]

Table 10: Fixed examples: first registered sensor of each modality, seed 42, 6000 G steps, test template 0. The sentence and both rule mappings are shown regardless of success. Responses longer than 280 characters are visibly truncated here; the evidence retains complete output.

Sensor	Held-out instruction	Target	Response
IMU/uci-har/test/01155	Complete this report with the observed category: The activity is CATEGORY.	The activity is walking.	The activity is walking.
IMU/uci-har/test/01155	For this sensor reading, map category walking to yes and all remaining categories to no. Return the mapped word.	yes	yes
IMU/uci-har/test/01155	For this sensor reading, map category walking to no and all remaining categories to yes. Return the mapped word.	no	yes
Audio/5-103415-A-2.wav	Complete this report with the observed category: The sound is CATEGORY.	The sound is pig.	!!!!

Table 11: All three U modalities composed at 1500 constituent steps. Negative changes from frozen improve BPB. Positive nonadditivity means combined BPB is worse than the prediction from summing standalone BPB improvements. We use that prediction as a descriptive reference. The analysis includes all subsets and both budgets.

Host	Operator	BPB [95%]	Δ frozen [95%]	Nonadditivity [95%]
Mini	Update mean	1.945 [1.927, 1.963]	-0.009 [-0.010, -0.009]	0.010 [0.009, 0.011]
Mini	Update sum	2.000 [1.981, 2.018]	0.046 [0.043, 0.048]	0.065 [0.063, 0.066]
Mini	Factor mean	1.949 [1.931, 1.967]	-0.005 [-0.005, -0.005]	0.014 [0.013, 0.016]
LFM 230M	Update mean	1.059 [1.044, 1.075]	-0.085 [-0.088, -0.083]	0.145 [0.141, 0.150]
LFM 230M	Update sum	1.193 [1.177, 1.210]	0.049 [0.044, 0.053]	0.279 [0.273, 0.286]
LFM 230M	Factor mean	1.070 [1.055, 1.086]	-0.074 [-0.077, -0.072]	0.156 [0.151, 0.162]
Qwen 0.8B	Update mean	0.965 [0.951, 0.979]	-0.033 [-0.035, -0.031]	0.009 [0.005, 0.013]
Qwen 0.8B	Update sum	1.103 [1.088, 1.118]	0.105 [0.100, 0.110]	0.147 [0.142, 0.152]
Qwen 0.8B	Factor mean	0.966 [0.951, 0.980]	-0.032 [-0.034, -0.031]	0.010 [0.005, 0.014]

time. Concurrent workers overlap, so the sums are neither isolated GPU-hours nor total study time. The measurements include the dense output head's cost.

Table 12: Resource measurements at batch size one on the isolated RTX PRO 6000. Parameters are millions of trainable adapter parameters. VRAM is peak allocated GiB with the host, adapter, and external encoder resident. First is median time from raw sensor input to the first token, in milliseconds; 32-token is median generation time with cached sensor factors and a fresh text cache. Both use 128 input text tokens. Frozen/static paths skip unused sensor work. The appendix contains every modality and prefix case.

Host	Task	Interface	Params M	VRAM GiB	First ms	32-token ms
Mini	IMU	Bridge	84.33	0.49	3.7	90.0
Mini	IMU	None	0.00	0.17	1.8	49.8
Mini	IMU	Static	0.33	0.17	2.7	82.3
LFM 230M	IMU	Bridge	199.07	1.67	6.5	161.0
LFM 230M	IMU	None	0.00	0.93	4.2	113.4
LFM 230M	IMU	Static	0.77	0.93	5.7	158.6
Qwen 0.8B	IMU	Bridge	410.66	4.52	50.6	440.1
Qwen 0.8B	IMU	None	0.00	2.99	46.8	359.8
Qwen 0.8B	IMU	Static	1.60	2.99	49.4	423.0

Timings at batch size one use one isolated RTX PRO 6000, 128 input text tokens, and 32 generated tokens. Raw inputs are already decoded in memory. The artifact separates preprocessing, encoding, adapter generation, prefill, and generation, and distinguishes a fresh sensor from cached factors. It records the device power limit, temperature, and clocks around each case. Native LFM/Qwen decoding uses a new text cache for each request; the released mini implementation recomputes its prefix. Variation across repeats describes this setup only. We did not measure variation across devices or applications, or efficiency on edge devices.

5 Related work

Hypernetworks generate the parameters of another network [3]. Adapters [4], LoRA [5], shared hypernetwork adapters [12], and HyperTuning [15] motivate adapting frozen hosts through smaller trainable interfaces. The bridge studied here conditions generated LoRA factors on continuous external encoder features. Its dense output head is large; freezing the host alone does not ensure parameter efficiency. Prompt tuning becomes more competitive with scale in the T5 experiments of Lester et al. [8]; our fixed input-prefix comparison is narrower in architecture and optimization.

The earlier 230M companion note (Paper 2 in this series) used the historical protocol [13]. This revision adds LFM comparisons with corrected evaluation and seed coverage; the companion's artifacts remain separate. The compression study (Paper 3) replaces the dense output head with coefficients over a learned or frozen basis of LoRA directions [14]. We retain the dense head here to measure its cost and behavior under the shared recipe. The compression study's results describe its separate architecture and protocol.

Multimodal systems such as Flamingo [1] and LLaVA [11] study broader visual-language behavior with different interfaces and supervision. Our closed-set ranking and deterministic response tasks cover a smaller range of behavior. Weight-space composition is related to task arithmetic [6]. For separately generated low-rank factors, averaging the factors differs from adding their effective updates.

6 Limitations and interpretation

The modular bridge is expensive and still requires paired examples for P and G. A new sensor task still needs supervision. The external encoders already provide strong label information, making a linear classifier and a label-in-text baseline necessary practical comparisons. We tested label ranking and short generated responses. These results leave open-ended descriptions, compositional cross-modal reasoning, and robust real-world sensor use unproven.

All hosts remain below one billion nominal language-model parameters and differ in several properties at once. Using one optimizer recipe makes the replication consistent but does not find each interface's best performance. The likelihood comparison applies to unpaired text; paired objectives may benefit from sensor use. The prefix results cover the tested input embeddings, leaving other soft prompts and per-layer prefixes untested. Variation in adapter geometry alone cannot show that the model uses sensors correctly.

The audio test set influenced encoder normalization, and the original encoder training records are incomplete. Excluding sources shared across folds addresses only that source overlap. Generalization is also limited by fixed test subjects, one audio holdout fold, image-domain transfer, and unknown host pretraining exposure. We report negative outcomes and every declared budget without choosing a checkpoint or penalty from test accuracy.

7 Reproducibility and revision record

The public reproduction package is `https://github.com/kortexa-ai/legolm.paper/tree/main/paper1`. All papers remain on the main branch with explicit per-paper entry points. The package has its own namespace and dependency lock. It includes the exact mini tokenizer, pinned input identities, all recipes, checkpoint hashes, per-item evaluation evidence, and an artifact-only verification command. Model execution and artifact verification are separate documented workflows. The renderer derives the primary tables, figures, and result text from the verified summary.

The normalization observations from the fixed batch are in the generation diagnostic artifact. Exact FP32 adapter checkpoints have a separate download index with immutable URLs, sizes, and hashes. The same index supplies the exact frozen feature caches, so encoder arithmetic on another device need not change a checkpoint's input features. Checkpoints can be evaluated without repeating training. Training them again requires the documented input assets and compute.

The initial decoder for native-chat examples stopped only at the model EOS and continued past valid assistant endings. Replaying saved checkpoints reproduced the repeated answers. We corrected the stopping rule, rescored the saved checkpoints, and excluded scores produced by the earlier decoder.

The original root-level `arxiv/` and `src/` remain the historical release. This revision reports new primary estimates from the corrected, registered matrix. Changes include matched prefix text budgets, separation of training and test corruption, native tokenization at label boundaries, full-population evaluation, and fresh larger-host runs. The changed evaluation protocol prevents pooling these results with historical scores. The matrix identifier is `1942c97abc73`. The evidence manifest SHA-256 is recorded in `release.json`, along with the exact summary and rendering sources. The source archive includes its bibliography and is compiled in a fresh directory before release. The package includes the original mini tokenizer, without retraining it.

References

[1] Jean-Baptiste Alayrac, Jeff Donahue, Pauline Luc, Antoine Miech, Iain Barr, Yana Hasson, Karel Lenc, Arthur Mensch, Katie Millican, Malcolm Reynolds, Roman Ring, Eliza Rutherford, Serkan Cabi, Tengda Han, Zhitao Gong, Sina Samangooei, Marianne Monteiro, Jacob Menick, Sebastian Borgeaud, Andrew Brock, Aida Nematzadeh, Sahand Sharifzadeh, Mikolaj Binkowski, Ricardo Barreira, Oriol Vinyals, Andrew Zisserman, and Karen Simonyan. Flamingo: a visual language model for few-shot learning. In *Advances in Neural Information Processing Systems*, 2022. arXiv:2204.14198.

[2] Davide Anguita, Alessandro Ghio, Luca Oneto, Xavier Parra, and Jorge Luis Reyes-Ortiz. A public domain dataset for human activity recognition using smartphones. In *European Symposium on Artificial Neural Networks, Computational Intelligence and Machine Learning*, 2013.

[3] David Ha, Andrew Dai, and Quoc V. Le. HyperNetworks. In *International Conference on Learning Representations*, 2017. arXiv:1609.09106.

[4] Neil Houlsby, Andrei Giurgiu, Stanislaw Jastrzebski, Bruna Morrone, Quentin de Laroussilhe, Andrea Gesmundo, Mona Attariyan, and Sylvain Gelly. Parameter-efficient transfer learning for NLP. In *International Conference on Machine Learning*, 2019. arXiv:1902.00751.

[5] Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022. arXiv:2106.09685.

[6] Gabriel Ilharco, Marco Tulio Ribeiro, Mitchell Wortsman, Suchin Gururangan, Ludwig Schmidt, Hannaneh Hajishirzi, and Ali Farhadi. Editing models with task arithmetic. In *International Conference on Learning Representations*, 2023. arXiv:2212.04089.

[7] Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. Dataset description: https://www.cs.toronto.edu/~kriz/cifar.html.

[8] Brian Lester, Rami Al-Rfou, and Noah Constant. The power of scale for parameter-efficient prompt tuning. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, pages 3045–3059, 2021.

[9] Xiang Lisa Li and Percy Liang. Prefix-Tuning: Optimizing continuous prompts for generation. In *Annual Meeting of the Association for Computational Linguistics*, 2021. arXiv:2101.00190.

[10] Liquid AI. LFM2.5-230M model card. https://huggingface.co/LiquidAI/LFM2.5-230M, 2026. Evaluated revision 40cb2ad3b3044d5a41eee083a6103c8b523afa45.

[11] Haotian Liu, Chunyuan Li, Qingyang Wu, and Yong Jae Lee. Visual instruction tuning. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2304.08485.

[12] Rabeeh Karimi Mahabadi, Sebastian Ruder, Mostafa Dehghani, and James Henderson. Parameter-efficient multi-task fine-tuning for transformers via shared hypernetworks. In *Annual Meeting of the Association for Computational Linguistics*, 2021. arXiv:2106.04489.

[13] Franci Penov. Conditional LoRA bridges at 230M: What replicates, what softens, and what inverts. Research note. https://raw.githubusercontent.com/kortexa-ai/legolm.paper/main/note/main.pdf, 2026. LegoLM Paper 2, July 3.

[14] Franci Penov. Two regimes of compressed conditioning: Basis-mixture bridges for frozen language models. Preprint. https://cdn.jsdelivr.net/gh/kortexa-ai/legolm.basis@main/arxiv2/main.pdf, 2026. LegoLM Paper 3, July 3.

[15] Jason Phang, Yi Mao, Pengcheng He, and Weizhu Chen. HyperTuning: Toward adapting large language models without back-propagation. In *International Conference on Machine Learning*, 2023. arXiv:2211.12485.

[16] Karol J. Piczak. ESC: Dataset for environmental sound classification. In *Proceedings of the 23rd ACM International Conference on Multimedia*, 2015.

[17] Qwen Team. Qwen3.5-0.8B model card. https://huggingface.co/Qwen/Qwen3.5-0.8B, 2026. Evaluated revision 2fc06364715b967f1860aea9cf38778875588b17.

A Complete registered comparisons

The machine-readable summary contains every condition, seed estimate, paired contrast, intervention, and intermediate checkpoint. These tables cover both primary endpoints and all declared penalties, including numerical failures. For an unavailable aggregate, $n = a/b$ gives available/declared seeds. We omit the aggregate when any declared seed is missing.

Table 13: Numerical failures with the specified training settings. Step ranges come from the original execution log. Diagnostic replays are separate from the training trials. The evidence includes checkpoints saved before failure.

Host/cohort	Task	Interface	Seed	λ	Failure step	Unavailable steps
Mini/P	IMU	bridge/constant	1042	0	2853	6000
Mini/P	IMU	bridge/constant	2042	0	5088	6000
Mini/P	IMU	bridge/constant	3042	0	3031	6000
Mini/P	IMU	bridge/constant	42	0	5009	6000
Mini/P	IMU	bridge/shuffled	1042	0	3295	6000
Mini/P	IMU	bridge/shuffled	2042	0	2520	6000
Mini/P	IMU	bridge/shuffled	42	0	3211	6000
Mini/P	Vision	bridge/shuffled	42	0	4347	6000
Mini/P	IMU	bridge/true	1042	0	1263	2400, 6000
Mini/P	IMU	bridge/true	2042	0	1170	1200, 2400, 6000
Mini/P	IMU	bridge/true	3042	0	1069	1200, 2400, 6000
Mini/P	IMU	bridge/true	4042	0	1823	2400, 6000
Mini/P	IMU	bridge/true	42	0	1301–1350	2400, 6000
Mini/P	IMU	bridge/true	1042	0.01	2211	2400, 6000
Mini/P	IMU	bridge/true	2042	0.01	2143	2400, 6000
Mini/P	IMU	bridge/true	42	0.01	2580	6000
Mini/P	IMU	bridge/true	1042	0.05	1388	2400, 6000
Mini/P	IMU	bridge/true	2042	0.05	5112	6000
Mini/P	IMU	bridge/true	42	0.05	545	600, 1200, 2400, 6000
Mini/P	IMU	bridge/true	1042	0.1	889	1200, 2400, 6000
Mini/P	IMU	bridge/true	2042	0.1	4007	6000
Mini/P	IMU	bridge/true	42	0.1	1698	2400, 6000
Mini/P	IMU	bridge/true	1042	0.2	807	1200, 2400, 6000
Mini/P	IMU	bridge/true	2042	0.2	2936	6000
Mini/P	IMU	bridge/true	42	0.2	1118	1200, 2400, 6000

Table 14: Native BPB; step/row count is shown explicitly. n is the number of training seeds; a frozen host is a fixed reference.

Host	Task	Interface	λ	Step/rows	n	BPB [95%]	Seed SD
LFM 230M	Audio	constant	0	3000/1024	3	1.067 [1.052, 1.083]	1.2×10^{-4}
LFM 230M	Audio	constant	0	300/256	3	1.098 [1.066, 1.131]	0.001
LFM 230M	IMU	constant	0	3000/1024	3	1.067 [1.052, 1.083]	0.001
LFM 230M	IMU	constant	0	300/256	3	1.098 [1.065, 1.130]	0.001
LFM 230M	Vision	constant	0	3000/1024	3	1.067 [1.052, 1.083]	4.3×10^{-4}
LFM 230M	Vision	constant	0	300/256	3	1.099 [1.066, 1.131]	0.002
Mini	Audio	constant	0	3000/1024	3	1.942 [1.924, 1.960]	2.5×10^{-4}
Mini	Audio	constant	0	300/256	3	1.963 [1.926, 1.999]	4.3×10^{-4}

Host	Task	Interface	λ	Step/rows	n	BPB [95%]	Seed SD
Mini	IMU	constant	0	3000/1024	3	1.943 [1.925, 1.961]	3.3×10^{-4}
Mini	IMU	constant	0	300/256	3	1.963 [1.926, 2.000]	3.3×10^{-4}
Mini	Vision	constant	0	3000/1024	3	1.942 [1.924, 1.960]	0.001
Mini	Vision	constant	0	300/256	3	1.963 [1.926, 2.000]	3.3×10^{-4}
Qwen 0.8B	Audio	constant	0	3000/1024	3	0.985 [0.970, 0.999]	0.001
Qwen 0.8B	Audio	constant	0	300/256	3	1.010 [0.981, 1.039]	0.001
Qwen 0.8B	IMU	constant	0	3000/1024	3	0.986 [0.972, 1.000]	0.001
Qwen 0.8B	IMU	constant	0	300/256	3	1.011 [0.982, 1.040]	0.001
Qwen 0.8B	Vision	constant	0	3000/1024	3	0.985 [0.971, 0.999]	0.001
Qwen 0.8B	Vision	constant	0	300/256	3	1.011 [0.982, 1.039]	0.001
LFM 230M	Audio	random	0	3000/1024	3	1.068 [1.052, 1.084]	4.1×10^{-4}
LFM 230M	Audio	random	0	300/256	3	1.102 [1.070, 1.135]	4.7×10^{-4}
LFM 230M	IMU	random	0	3000/1024	3	1.067 [1.052, 1.083]	3.4×10^{-4}
LFM 230M	IMU	random	0	300/256	3	1.102 [1.070, 1.134]	0.001
LFM 230M	Vision	random	0	3000/1024	3	1.068 [1.052, 1.084]	4.1×10^{-4}
LFM 230M	Vision	random	0	300/256	3	1.102 [1.070, 1.135]	4.7×10^{-4}
Mini	Audio	random	0	3000/1024	3	1.943 [1.925, 1.961]	2.6×10^{-4}
Mini	Audio	random	0	300/256	3	1.965 [1.928, 2.001]	2.5×10^{-4}
Mini	IMU	random	0	3000/1024	3	1.943 [1.925, 1.961]	4.8×10^{-4}
Mini	IMU	random	0	300/256	3	1.964 [1.927, 2.001]	3.3×10^{-4}
Mini	Vision	random	0	3000/1024	3	1.943 [1.925, 1.961]	2.6×10^{-4}
Mini	Vision	random	0	300/256	3	1.965 [1.928, 2.001]	2.5×10^{-4}
Qwen 0.8B	Audio	random	0	3000/1024	3	0.987 [0.973, 1.001]	8.9×10^{-5}
Qwen 0.8B	Audio	random	0	300/256	3	1.013 [0.984, 1.042]	0.001
Qwen 0.8B	IMU	random	0	3000/1024	3	0.986 [0.972, 1.000]	0.001
Qwen 0.8B	IMU	random	0	300/256	3	1.014 [0.985, 1.042]	0.002
Qwen 0.8B	Vision	random	0	3000/1024	3	0.987 [0.973, 1.001]	8.9×10^{-5}
Qwen 0.8B	Vision	random	0	300/256	3	1.013 [0.984, 1.042]	0.001
LFM 230M	Audio	shuffled	0	3000/1024	3	1.068 [1.052, 1.084]	0.001
LFM 230M	Audio	shuffled	0	300/256	3	1.099 [1.066, 1.131]	0.001
LFM 230M	IMU	shuffled	0	3000/1024	3	1.068 [1.052, 1.084]	3.6×10^{-4}
LFM 230M	IMU	shuffled	0	300/256	3	1.099 [1.066, 1.131]	0.001
LFM 230M	Vision	shuffled	0	3000/1024	3	1.068 [1.052, 1.084]	0.001
LFM 230M	Vision	shuffled	0	300/256	3	1.098 [1.066, 1.130]	4.5×10^{-4}
Mini	Audio	shuffled	0	3000/1024	3	1.942 [1.924, 1.960]	1.3×10^{-4}
Mini	Audio	shuffled	0	300/256	3	1.963 [1.926, 2.000]	2.1×10^{-4}
Mini	IMU	shuffled	0	3000/1024	3	1.942 [1.924, 1.960]	2.3×10^{-4}
Mini	IMU	shuffled	0	300/256	3	1.963 [1.926, 2.000]	1.3×10^{-4}
Mini	Vision	shuffled	0	3000/1024	3	1.942 [1.924, 1.960]	3.5×10^{-4}
Mini	Vision	shuffled	0	300/256	3	1.963 [1.926, 1.999]	2.2×10^{-4}
Qwen 0.8B	Audio	shuffled	0	3000/1024	3	0.984 [0.970, 0.999]	2.7×10^{-4}
Qwen 0.8B	Audio	shuffled	0	300/256	3	1.010 [0.981, 1.039]	0.002
Qwen 0.8B	IMU	shuffled	0	3000/1024	3	0.985 [0.971, 1.000]	4.6×10^{-4}
Qwen 0.8B	IMU	shuffled	0	300/256	3	1.011 [0.982, 1.040]	0.002
Qwen 0.8B	Vision	shuffled	0	3000/1024	3	0.984 [0.970, 0.998]	0.001
Qwen 0.8B	Vision	shuffled	0	300/256	3	1.010 [0.981, 1.039]	0.001
LFM 230M	Audio	true	0	3000/1024	3	1.067 [1.051, 1.083]	0.001
LFM 230M	Audio	true	0	300/256	3	1.098 [1.066, 1.131]	4.0×10^{-4}
LFM 230M	IMU	true	0	3000/1024	3	1.068 [1.052, 1.084]	1.6×10^{-4}
LFM 230M	IMU	true	0	300/256	3	1.098 [1.065, 1.130]	0.001
LFM 230M	Vision	true	0	3000/1024	3	1.068 [1.052, 1.084]	3.1×10^{-4}
LFM 230M	Vision	true	0	300/256	3	1.099 [1.066, 1.131]	4.9×10^{-4}

Host	Task	Interface	λ	Step/rows	n	BPB [95%]	Seed SD
LFM 230M	Audio	Frozen	0	0/1024	1	1.145 [1.128, 1.161]	–
LFM 230M	Audio	Frozen	0	0/256	1	1.184 [1.150, 1.218]	–
LFM 230M	IMU	Frozen	0	0/1024	1	1.145 [1.128, 1.161]	–
LFM 230M	IMU	Frozen	0	0/256	1	1.184 [1.150, 1.218]	–
LFM 230M	Vision	Frozen	0	0/1024	1	1.145 [1.128, 1.161]	–
LFM 230M	Vision	Frozen	0	0/256	1	1.184 [1.150, 1.218]	–
LFM 230M	Audio	prefix	0	3000/1024	3	1.072 [1.057, 1.088]	1.1×10^{-4}
LFM 230M	Audio	prefix	0	0/256	3	1.198 [1.163, 1.232]	0.003
LFM 230M	Audio	prefix	0	300/256	3	1.114 [1.082, 1.146]	0.001
LFM 230M	IMU	prefix	0	3000/1024	3	1.070 [1.055, 1.086]	1.3×10^{-4}
LFM 230M	IMU	prefix	0	0/256	3	1.192 [1.158, 1.226]	0.005
LFM 230M	IMU	prefix	0	300/256	3	1.114 [1.082, 1.146]	0.001
LFM 230M	Vision	prefix	0	3000/1024	3	1.074 [1.058, 1.090]	4.7×10^{-4}
LFM 230M	Vision	prefix	0	0/256	3	1.192 [1.158, 1.225]	0.001
LFM 230M	Vision	prefix	0	300/256	3	1.115 [1.083, 1.147]	3.9×10^{-4}
LFM 230M	Audio	static	0	3000/1024	3	1.066 [1.051, 1.083]	1.4×10^{-4}
LFM 230M	Audio	static	0	300/256	3	1.096 [1.064, 1.129]	9.9×10^{-5}
LFM 230M	IMU	static	0	3000/1024	3	1.066 [1.051, 1.083]	1.4×10^{-4}
LFM 230M	IMU	static	0	300/256	3	1.096 [1.064, 1.129]	9.9×10^{-5}
LFM 230M	Vision	static	0	3000/1024	3	1.066 [1.051, 1.083]	1.4×10^{-4}
LFM 230M	Vision	static	0	300/256	3	1.096 [1.064, 1.129]	9.9×10^{-5}
Mini	Audio	true	0	3000/1024	3	1.942 [1.924, 1.960]	0.001
Mini	Audio	true	0	300/256	3	1.963 [1.926, 1.999]	1.9×10^{-4}
Mini	IMU	true	0	3000/1024	3	1.943 [1.925, 1.961]	1.7×10^{-4}
Mini	IMU	true	0	300/256	3	1.963 [1.926, 1.999]	3.4×10^{-4}
Mini	Vision	true	0	3000/1024	3	1.942 [1.924, 1.960]	3.1×10^{-4}
Mini	Vision	true	0	300/256	3	1.963 [1.926, 1.999]	7.9×10^{-5}
Mini	Audio	Frozen	0	0/1024	1	1.954 [1.936, 1.973]	–
Mini	Audio	Frozen	0	0/256	1	1.965 [1.929, 2.004]	–
Mini	IMU	Frozen	0	0/1024	1	1.954 [1.936, 1.973]	–
Mini	IMU	Frozen	0	0/256	1	1.965 [1.929, 2.004]	–
Mini	Vision	Frozen	0	0/1024	1	1.954 [1.936, 1.973]	–
Mini	Vision	Frozen	0	0/256	1	1.965 [1.929, 2.004]	–
Mini	Audio	prefix	0	3000/1024	3	1.953 [1.934, 1.971]	4.8×10^{-5}
Mini	Audio	prefix	0	0/256	3	1.965 [1.928, 2.002]	1.5×10^{-4}
Mini	Audio	prefix	0	300/256	3	1.964 [1.927, 2.001]	6.7×10^{-5}
Mini	IMU	prefix	0	3000/1024	3	1.953 [1.934, 1.971]	3.5×10^{-5}
Mini	IMU	prefix	0	0/256	3	1.965 [1.928, 2.002]	2.0×10^{-4}
Mini	IMU	prefix	0	300/256	3	1.965 [1.928, 2.001]	2.6×10^{-4}
Mini	Vision	prefix	0	3000/1024	3	1.953 [1.934, 1.971]	2.9×10^{-5}
Mini	Vision	prefix	0	0/256	3	1.965 [1.929, 2.002]	8.4×10^{-5}
Mini	Vision	prefix	0	300/256	3	1.964 [1.927, 2.001]	4.4×10^{-6}
Mini	Audio	static	0	3000/1024	3	1.941 [1.923, 1.959]	1.4×10^{-4}
Mini	Audio	static	0	300/256	3	1.962 [1.925, 1.998]	1.5×10^{-4}
Mini	IMU	static	0	3000/1024	3	1.941 [1.923, 1.959]	1.4×10^{-4}
Mini	IMU	static	0	300/256	3	1.962 [1.925, 1.998]	1.5×10^{-4}
Mini	Vision	static	0	3000/1024	3	1.941 [1.923, 1.959]	1.4×10^{-4}
Mini	Vision	static	0	300/256	3	1.962 [1.925, 1.998]	1.5×10^{-4}
Qwen 0.8B	Audio	true	0	3000/1024	3	0.985 [0.971, 0.999]	1.7×10^{-4}
Qwen 0.8B	Audio	true	0	300/256	3	1.011 [0.982, 1.040]	0.001
Qwen 0.8B	IMU	true	0	3000/1024	3	0.986 [0.972, 1.001]	3.5×10^{-4}
Qwen 0.8B	IMU	true	0	300/256	3	1.010 [0.981, 1.039]	0.001

Host	Task	Interface	λ	Step/rows	n	BPB [95%]	Seed SD
Qwen 0.8B	Vision	true	0	3000/1024	3	0.984 [0.970, 0.998]	0.001
Qwen 0.8B	Vision	true	0	300/256	3	1.010 [0.981, 1.039]	0.001
Qwen 0.8B	Audio	Frozen	0	0/1024	1	0.998 [0.984, 1.012]	–
Qwen 0.8B	Audio	Frozen	0	0/256	1	1.040 [1.010, 1.069]	–
Qwen 0.8B	IMU	Frozen	0	0/1024	1	0.998 [0.984, 1.012]	–
Qwen 0.8B	IMU	Frozen	0	0/256	1	1.040 [1.010, 1.069]	–
Qwen 0.8B	Vision	Frozen	0	0/1024	1	0.998 [0.984, 1.012]	–
Qwen 0.8B	Vision	Frozen	0	0/256	1	1.040 [1.010, 1.069]	–
Qwen 0.8B	Audio	prefix	0	3000/1024	3	0.963 [0.949, 0.977]	4.1×10^{-4}
Qwen 0.8B	Audio	prefix	0	0/256	3	1.280 [1.219, 1.347]	0.058
Qwen 0.8B	Audio	prefix	0	300/256	3	1.004 [0.975, 1.033]	2.5×10^{-4}
Qwen 0.8B	IMU	prefix	0	3000/1024	3	0.962 [0.948, 0.976]	2.7×10^{-4}
Qwen 0.8B	IMU	prefix	0	0/256	3	1.299 [1.253, 1.344]	0.026
Qwen 0.8B	IMU	prefix	0	300/256	3	1.004 [0.975, 1.033]	0.001
Qwen 0.8B	Vision	prefix	0	3000/1024	3	0.964 [0.949, 0.978]	0.001
Qwen 0.8B	Vision	prefix	0	0/256	3	1.358 [1.303, 1.414]	0.043
Qwen 0.8B	Vision	prefix	0	300/256	3	1.004 [0.976, 1.033]	0.001
Qwen 0.8B	Audio	static	0	3000/1024	3	0.987 [0.972, 1.001]	0.001
Qwen 0.8B	Audio	static	0	300/256	3	1.009 [0.979, 1.037]	2.3×10^{-4}
Qwen 0.8B	IMU	static	0	3000/1024	3	0.987 [0.972, 1.001]	0.001
Qwen 0.8B	IMU	static	0	300/256	3	1.009 [0.979, 1.037]	2.3×10^{-4}
Qwen 0.8B	Vision	static	0	3000/1024	3	0.987 [0.972, 1.001]	0.001
Qwen 0.8B	Vision	static	0	300/256	3	1.009 [0.979, 1.037]	2.3×10^{-4}
LFM 230M	IMU	true	0.01	3000/1024	3	1.068 [1.052, 1.084]	3.4×10^{-4}
LFM 230M	IMU	true	0.01	300/256	3	1.099 [1.067, 1.131]	0.001
Mini	IMU	true	0.01	3000/1024	3	1.943 [1.925, 1.961]	3.5×10^{-4}
Mini	IMU	true	0.01	300/256	3	1.963 [1.926, 2.000]	3.6×10^{-4}
Qwen 0.8B	IMU	true	0.01	3000/1024	3	0.985 [0.971, 0.999]	3.8×10^{-4}
Qwen 0.8B	IMU	true	0.01	300/256	3	1.010 [0.980, 1.039]	0.002
LFM 230M	IMU	true	0.05	3000/1024	3	1.068 [1.052, 1.084]	2.1×10^{-4}
LFM 230M	IMU	true	0.05	300/256	3	1.099 [1.067, 1.131]	3.6×10^{-4}
Mini	IMU	true	0.05	3000/1024	3	1.949 [1.931, 1.967]	4.9×10^{-4}
Mini	IMU	true	0.05	300/256	3	1.965 [1.928, 2.001]	0.001
Qwen 0.8B	IMU	true	0.05	3000/1024	3	0.985 [0.971, 0.999]	0.001
Qwen 0.8B	IMU	true	0.05	300/256	3	1.010 [0.981, 1.039]	0.002
LFM 230M	IMU	true	0.1	3000/1024	3	1.068 [1.053, 1.084]	0.001
LFM 230M	IMU	true	0.1	300/256	3	1.106 [1.074, 1.138]	0.002
Mini	IMU	true	0.1	3000/1024	3	1.949 [1.931, 1.968]	2.2×10^{-4}
Mini	IMU	true	0.1	300/256	3	1.965 [1.928, 2.002]	0.001
Qwen 0.8B	IMU	true	0.1	3000/1024	3	0.986 [0.972, 1.000]	2.4×10^{-4}
Qwen 0.8B	IMU	true	0.1	300/256	3	1.009 [0.979, 1.037]	0.001
LFM 230M	IMU	true	0.2	3000/1024	3	1.072 [1.057, 1.088]	3.3×10^{-4}
LFM 230M	IMU	true	0.2	300/256	3	1.111 [1.078, 1.143]	0.001
Mini	IMU	true	0.2	3000/1024	3	1.949 [1.931, 1.967]	0.001
Mini	IMU	true	0.2	300/256	3	1.966 [1.929, 2.002]	0.001
Qwen 0.8B	IMU	true	0.2	3000/1024	3	0.988 [0.974, 1.002]	0.001
Qwen 0.8B	IMU	true	0.2	300/256	3	1.017 [0.988, 1.047]	0.002

Table 15: Common-byte BPB on identical held-out spans. n is the number of training seeds; a frozen host is a fixed reference.

Host	Task	Interface	λ	Step	n	BPB [95%]	Seed SD
LFM 230M	Audio	constant	0	3000	3	1.268 [1.237, 1.300]	0.001
LFM 230M	Audio	constant	0	300	3	1.271 [1.240, 1.302]	0.001
LFM 230M	IMU	constant	0	3000	3	1.268 [1.238, 1.300]	0.001
LFM 230M	IMU	constant	0	300	3	1.272 [1.241, 1.304]	0.002
LFM 230M	Vision	constant	0	3000	3	1.270 [1.239, 1.301]	4.3×10^{-4}
LFM 230M	Vision	constant	0	300	3	1.271 [1.240, 1.302]	0.002
Mini	Audio	constant	0	3000	3	2.122 [2.082, 2.164]	3.6×10^{-4}
Mini	Audio	constant	0	300	3	2.139 [2.098, 2.182]	2.8×10^{-4}
Mini	IMU	constant	0	3000	3	2.124 [2.083, 2.165]	4.2×10^{-4}
Mini	IMU	constant	0	300	3	2.140 [2.099, 2.182]	2.5×10^{-4}
Mini	Vision	constant	0	3000	3	2.122 [2.081, 2.164]	1.6×10^{-4}
Mini	Vision	constant	0	300	3	2.139 [2.098, 2.181]	4.8×10^{-4}
Qwen 0.8B	Audio	constant	0	3000	3	1.189 [1.161, 1.217]	3.7×10^{-4}
Qwen 0.8B	Audio	constant	0	300	3	1.179 [1.151, 1.206]	0.001
Qwen 0.8B	IMU	constant	0	3000	3	1.187 [1.160, 1.216]	0.001
Qwen 0.8B	IMU	constant	0	300	3	1.183 [1.155, 1.210]	0.002
Qwen 0.8B	Vision	constant	0	3000	3	1.187 [1.159, 1.216]	0.001
Qwen 0.8B	Vision	constant	0	300	3	1.180 [1.153, 1.207]	0.001
LFM 230M	Audio	random	0	3000	3	1.270 [1.239, 1.301]	4.1×10^{-4}
LFM 230M	Audio	random	0	300	3	1.276 [1.245, 1.307]	0.001
LFM 230M	IMU	random	0	3000	3	1.269 [1.238, 1.301]	0.002
LFM 230M	IMU	random	0	300	3	1.276 [1.245, 1.307]	4.6×10^{-4}
LFM 230M	Vision	random	0	3000	3	1.270 [1.239, 1.301]	4.1×10^{-4}
LFM 230M	Vision	random	0	300	3	1.276 [1.245, 1.307]	0.001
Mini	Audio	random	0	3000	3	2.125 [2.084, 2.166]	9.0×10^{-5}
Mini	Audio	random	0	300	3	2.141 [2.101, 2.184]	0.001
Mini	IMU	random	0	3000	3	2.125 [2.084, 2.166]	3.3×10^{-4}
Mini	IMU	random	0	300	3	2.141 [2.100, 2.183]	0.001
Mini	Vision	random	0	3000	3	2.125 [2.084, 2.166]	9.0×10^{-5}
Mini	Vision	random	0	300	3	2.141 [2.101, 2.184]	0.001
Qwen 0.8B	Audio	random	0	3000	3	1.190 [1.162, 1.219]	0.002
Qwen 0.8B	Audio	random	0	300	3	1.182 [1.155, 1.210]	0.001
Qwen 0.8B	IMU	random	0	3000	3	1.189 [1.161, 1.217]	0.003
Qwen 0.8B	IMU	random	0	300	3	1.184 [1.156, 1.212]	0.001
Qwen 0.8B	Vision	random	0	3000	3	1.190 [1.162, 1.219]	0.002
Qwen 0.8B	Vision	random	0	300	3	1.182 [1.155, 1.210]	0.001
LFM 230M	Audio	shuffled	0	3000	3	1.269 [1.238, 1.300]	0.002
LFM 230M	Audio	shuffled	0	300	3	1.272 [1.240, 1.303]	4.4×10^{-4}
LFM 230M	IMU	shuffled	0	3000	3	1.270 [1.239, 1.301]	0.001
LFM 230M	IMU	shuffled	0	300	3	1.271 [1.240, 1.302]	0.001
LFM 230M	Vision	shuffled	0	3000	3	1.268 [1.237, 1.299]	0.002
LFM 230M	Vision	shuffled	0	300	3	1.270 [1.239, 1.301]	4.5×10^{-4}
Mini	Audio	shuffled	0	3000	3	2.122 [2.082, 2.164]	0.001
Mini	Audio	shuffled	0	300	3	2.140 [2.099, 2.182]	0.001
Mini	IMU	shuffled	0	3000	3	2.124 [2.083, 2.165]	3.7×10^{-4}
Mini	IMU	shuffled	0	300	3	2.139 [2.098, 2.181]	1.7×10^{-4}
Mini	Vision	shuffled	0	3000	3	2.122 [2.082, 2.164]	0.001
Mini	Vision	shuffled	0	300	3	2.139 [2.098, 2.181]	2.4×10^{-4}
Qwen 0.8B	Audio	shuffled	0	3000	3	1.187 [1.159, 1.216]	0.002

Host	Task	Interface	λ	Step	n	BPB [95%]	Seed SD
Qwen 0.8B	Audio	shuffled	0	300	3	1.181 [1.153, 1.209]	0.004
Qwen 0.8B	IMU	shuffled	0	3000	3	1.187 [1.159, 1.216]	0.002
Qwen 0.8B	IMU	shuffled	0	300	3	1.181 [1.154, 1.209]	0.001
Qwen 0.8B	Vision	shuffled	0	3000	3	1.187 [1.159, 1.215]	0.001
Qwen 0.8B	Vision	shuffled	0	300	3	1.180 [1.153, 1.208]	0.001
LFM 230M	Audio	true	0	3000	3	1.266 [1.235, 1.298]	0.001
LFM 230M	Audio	true	0	300	3	1.271 [1.240, 1.302]	3.4×10^{-4}
LFM 230M	IMU	true	0	3000	3	1.269 [1.238, 1.300]	0.002
LFM 230M	IMU	true	0	300	3	1.272 [1.241, 1.303]	0.001
LFM 230M	Vision	true	0	3000	3	1.269 [1.238, 1.300]	2.6×10^{-4}
LFM 230M	Vision	true	0	300	3	1.271 [1.240, 1.302]	0.002
LFM 230M	Audio	Frozen	0	0	1	1.373 [1.340, 1.407]	–
LFM 230M	IMU	Frozen	0	0	1	1.373 [1.340, 1.407]	–
LFM 230M	Vision	Frozen	0	0	1	1.373 [1.340, 1.407]	–
LFM 230M	Audio	prefix	0	0	3	1.385 [1.351, 1.419]	0.002
LFM 230M	Audio	prefix	0	3000	3	1.275 [1.243, 1.306]	4.9×10^{-4}
LFM 230M	Audio	prefix	0	300	3	1.280 [1.248, 1.312]	0.001
LFM 230M	IMU	prefix	0	0	3	1.382 [1.348, 1.416]	0.001
LFM 230M	IMU	prefix	0	3000	3	1.274 [1.242, 1.305]	2.8×10^{-4}
LFM 230M	IMU	prefix	0	300	3	1.282 [1.250, 1.314]	3.5×10^{-4}
LFM 230M	Vision	prefix	0	0	3	1.379 [1.345, 1.413]	0.004
LFM 230M	Vision	prefix	0	3000	3	1.276 [1.245, 1.308]	0.001
LFM 230M	Vision	prefix	0	300	3	1.282 [1.251, 1.315]	0.002
LFM 230M	Audio	static	0	3000	3	1.268 [1.237, 1.299]	0.001
LFM 230M	Audio	static	0	300	3	1.271 [1.240, 1.303]	2.9×10^{-4}
LFM 230M	IMU	static	0	3000	3	1.268 [1.237, 1.299]	0.001
LFM 230M	IMU	static	0	300	3	1.271 [1.240, 1.303]	2.9×10^{-4}
LFM 230M	Vision	static	0	3000	3	1.268 [1.237, 1.299]	0.001
LFM 230M	Vision	static	0	300	3	1.271 [1.240, 1.303]	2.9×10^{-4}
Mini	Audio	true	0	3000	3	2.122 [2.082, 2.164]	0.001
Mini	Audio	true	0	300	3	2.139 [2.098, 2.182]	3.3×10^{-4}
Mini	IMU	true	0	3000	3	2.123 [2.083, 2.165]	0.001
Mini	IMU	true	0	300	3	2.139 [2.098, 2.181]	4.1×10^{-4}
Mini	Vision	true	0	3000	3	2.123 [2.082, 2.165]	4.9×10^{-4}
Mini	Vision	true	0	300	3	2.139 [2.098, 2.181]	2.2×10^{-4}
Mini	Audio	Frozen	0	0	1	2.142 [2.103, 2.185]	–
Mini	IMU	Frozen	0	0	1	2.142 [2.103, 2.185]	–
Mini	Vision	Frozen	0	0	1	2.142 [2.103, 2.185]	–
Mini	Audio	prefix	0	0	3	2.144 [2.103, 2.186]	0.001
Mini	Audio	prefix	0	3000	3	2.140 [2.099, 2.183]	5.3×10^{-5}
Mini	Audio	prefix	0	300	3	2.141 [2.100, 2.184]	3.1×10^{-4}
Mini	IMU	prefix	0	0	3	2.144 [2.103, 2.186]	3.3×10^{-4}
Mini	IMU	prefix	0	3000	3	2.140 [2.099, 2.182]	7.5×10^{-5}
Mini	IMU	prefix	0	300	3	2.142 [2.100, 2.184]	4.0×10^{-4}
Mini	Vision	prefix	0	0	3	2.144 [2.103, 2.186]	0.001
Mini	Vision	prefix	0	3000	3	2.140 [2.099, 2.183]	6.7×10^{-5}
Mini	Vision	prefix	0	300	3	2.142 [2.100, 2.184]	1.7×10^{-4}
Mini	Audio	static	0	3000	3	2.122 [2.081, 2.164]	3.3×10^{-4}
Mini	Audio	static	0	300	3	2.138 [2.097, 2.180]	2.0×10^{-4}
Mini	IMU	static	0	3000	3	2.122 [2.081, 2.164]	3.3×10^{-4}
Mini	IMU	static	0	300	3	2.138 [2.097, 2.180]	2.0×10^{-4}
Mini	Vision	static	0	3000	3	2.122 [2.081, 2.164]	3.3×10^{-4}

Host	Task	Interface	λ	Step	n	BPB [95%]	Seed SD
Mini	Vision	static	0	300	3	2.138 [2.097, 2.180]	2.0×10^{-4}
Qwen 0.8B	Audio	true	0	3000	3	1.187 [1.159, 1.216]	0.001
Qwen 0.8B	Audio	true	0	300	3	1.180 [1.152, 1.207]	0.002
Qwen 0.8B	IMU	true	0	3000	3	1.189 [1.161, 1.218]	0.003
Qwen 0.8B	IMU	true	0	300	3	1.180 [1.152, 1.207]	0.001
Qwen 0.8B	Vision	true	0	3000	3	1.187 [1.159, 1.215]	0.002
Qwen 0.8B	Vision	true	0	300	3	1.180 [1.152, 1.208]	0.002
Qwen 0.8B	Audio	Frozen	0	0	1	1.212 [1.184, 1.241]	–
Qwen 0.8B	IMU	Frozen	0	0	1	1.212 [1.184, 1.241]	–
Qwen 0.8B	Vision	Frozen	0	0	1	1.212 [1.184, 1.241]	–
Qwen 0.8B	Audio	prefix	0	0	3	1.611 [1.535, 1.702]	0.084
Qwen 0.8B	Audio	prefix	0	3000	3	1.165 [1.137, 1.193]	0.001
Qwen 0.8B	Audio	prefix	0	300	3	1.173 [1.145, 1.201]	0.002
Qwen 0.8B	IMU	prefix	0	0	3	1.653 [1.577, 1.732]	0.076
Qwen 0.8B	IMU	prefix	0	3000	3	1.163 [1.136, 1.191]	2.6×10^{-4}
Qwen 0.8B	IMU	prefix	0	300	3	1.172 [1.144, 1.200]	0.001
Qwen 0.8B	Vision	prefix	0	0	3	1.743 [1.671, 1.819]	0.067
Qwen 0.8B	Vision	prefix	0	3000	3	1.166 [1.139, 1.194]	0.001
Qwen 0.8B	Vision	prefix	0	300	3	1.173 [1.145, 1.201]	0.001
Qwen 0.8B	Audio	static	0	3000	3	1.189 [1.162, 1.218]	0.001
Qwen 0.8B	Audio	static	0	300	3	1.178 [1.151, 1.205]	0.001
Qwen 0.8B	IMU	static	0	3000	3	1.189 [1.162, 1.218]	0.001
Qwen 0.8B	IMU	static	0	300	3	1.178 [1.151, 1.205]	0.001
Qwen 0.8B	Vision	static	0	3000	3	1.189 [1.162, 1.218]	0.001
Qwen 0.8B	Vision	static	0	300	3	1.178 [1.151, 1.205]	0.001
LFM 230M	IMU	true	0.01	3000	3	1.270 [1.239, 1.301]	0.001
LFM 230M	IMU	true	0.01	300	3	1.272 [1.241, 1.304]	0.001
Mini	IMU	true	0.01	3000	3	2.124 [2.084, 2.166]	0.001
Mini	IMU	true	0.01	300	3	2.140 [2.098, 2.182]	3.4×10^{-4}
Qwen 0.8B	IMU	true	0.01	3000	3	1.188 [1.160, 1.217]	0.001
Qwen 0.8B	IMU	true	0.01	300	3	1.180 [1.152, 1.207]	0.001
LFM 230M	IMU	true	0.05	3000	3	1.269 [1.238, 1.301]	0.002
LFM 230M	IMU	true	0.05	300	3	1.272 [1.241, 1.303]	0.001
Mini	IMU	true	0.05	3000	3	2.133 [2.092, 2.175]	0.001
Mini	IMU	true	0.05	300	3	2.142 [2.101, 2.184]	0.001
Qwen 0.8B	IMU	true	0.05	3000	3	1.188 [1.160, 1.217]	0.001
Qwen 0.8B	IMU	true	0.05	300	3	1.180 [1.153, 1.208]	0.002
LFM 230M	IMU	true	0.1	3000	3	1.269 [1.238, 1.301]	0.003
LFM 230M	IMU	true	0.1	300	3	1.280 [1.249, 1.311]	0.003
Mini	IMU	true	0.1	3000	3	2.134 [2.093, 2.176]	0.001
Mini	IMU	true	0.1	300	3	2.142 [2.101, 2.184]	1.9×10^{-4}
Qwen 0.8B	IMU	true	0.1	3000	3	1.191 [1.163, 1.219]	0.001
Qwen 0.8B	IMU	true	0.1	300	3	1.177 [1.150, 1.205]	0.002
LFM 230M	IMU	true	0.2	3000	3	1.276 [1.245, 1.307]	0.001
LFM 230M	IMU	true	0.2	300	3	1.284 [1.253, 1.316]	0.001
Mini	IMU	true	0.2	3000	3	2.132 [2.092, 2.175]	0.001
Mini	IMU	true	0.2	300	3	2.143 [2.102, 2.185]	0.001
Qwen 0.8B	IMU	true	0.2	3000	3	1.191 [1.163, 1.220]	0.004
Qwen 0.8B	IMU	true	0.2	300	3	1.188 [1.160, 1.216]	0.004

Table 16: Paired P ranking on the full held-out population. n is the number of training seeds; a frozen host is a fixed reference.

Host	Task	Interface	λ	Step	n	Macro [95%]	Micro [95%]
LFM 230M	Audio	constant	0	6000	3	0.018 [0.010, 0.022]	0.018 [0.010, 0.024]
LFM 230M	Audio	constant	0	600	3	0.018 [0.013, 0.022]	0.018 [0.013, 0.024]
LFM 230M	IMU	constant	0	6000	3	0.167 [0.167, 0.167]	0.148 [0.138, 0.159]
LFM 230M	IMU	constant	0	600	3	0.167 [0.167, 0.167]	0.160 [0.155, 0.164]
Mini	Audio	constant	0	6000	5	0.020 [0.017, 0.020]	0.020 [0.017, 0.023]
Mini	Audio	constant	0	600	5	0.024 [0.020, 0.029]	0.024 [0.019, 0.029]
Mini	IMU	constant	0	6000	1/5	–	–
Mini	IMU	constant	0	600	5	0.167 [0.167, 0.167]	0.156 [0.148, 0.163]
Mini	Vision	constant	0	6000	3	0.100 [0.100, 0.101]	0.100 [0.100, 0.101]
Mini	Vision	constant	0	600	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Audio	constant	0	6000	5	0.021 [0.018, 0.024]	0.021 [0.016, 0.026]
Qwen 0.8B	Audio	constant	0	600	5	0.017 [0.011, 0.022]	0.017 [0.010, 0.024]
Qwen 0.8B	IMU	constant	0	6000	5	0.167 [0.167, 0.167]	0.153 [0.144, 0.161]
Qwen 0.8B	IMU	constant	0	600	5	0.167 [0.166, 0.167]	0.153 [0.144, 0.161]
Qwen 0.8B	Vision	constant	0	6000	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Vision	constant	0	600	3	0.105 [0.100, 0.114]	0.105 [0.100, 0.114]
LFM 230M	Audio	random	0	6000	3	0.023 [0.020, 0.028]	0.023 [0.017, 0.030]
LFM 230M	Audio	random	0	600	3	0.024 [0.018, 0.031]	0.024 [0.017, 0.032]
LFM 230M	IMU	random	0	6000	3	0.109 [0.027, 0.167]	0.104 [0.024, 0.159]
LFM 230M	IMU	random	0	600	3	0.124 [0.045, 0.167]	0.113 [0.042, 0.159]
Mini	Audio	random	0	6000	5	0.021 [0.020, 0.023]	0.021 [0.018, 0.024]
Mini	Audio	random	0	600	5	0.021 [0.018, 0.025]	0.021 [0.017, 0.028]
Mini	IMU	random	0	6000	5	0.190 [0.158, 0.227]	0.179 [0.140, 0.221]
Mini	IMU	random	0	600	5	0.123 [0.069, 0.168]	0.112 [0.061, 0.157]
Mini	Vision	random	0	6000	3	0.112 [0.101, 0.122]	0.112 [0.101, 0.122]
Mini	Vision	random	0	600	3	0.084 [0.053, 0.100]	0.084 [0.053, 0.100]
Qwen 0.8B	Audio	random	0	6000	5	0.023 [0.020, 0.029]	0.023 [0.018, 0.031]
Qwen 0.8B	Audio	random	0	600	5	0.023 [0.015, 0.032]	0.023 [0.014, 0.033]
Qwen 0.8B	IMU	random	0	6000	5	0.175 [0.167, 0.194]	0.160 [0.145, 0.180]
Qwen 0.8B	IMU	random	0	600	5	0.209 [0.153, 0.269]	0.199 [0.145, 0.259]
Qwen 0.8B	Vision	random	0	6000	3	0.100 [0.099, 0.100]	0.100 [0.099, 0.100]
Qwen 0.8B	Vision	random	0	600	3	0.093 [0.066, 0.113]	0.093 [0.066, 0.113]
LFM 230M	Audio	shuffled	0	6000	3	0.020 [0.020, 0.020]	0.020 [0.017, 0.025]
LFM 230M	Audio	shuffled	0	600	3	0.021 [0.016, 0.026]	0.021 [0.015, 0.028]
LFM 230M	IMU	shuffled	0	6000	3	0.176 [0.167, 0.196]	0.164 [0.155, 0.177]
LFM 230M	IMU	shuffled	0	600	3	0.156 [0.129, 0.167]	0.145 [0.114, 0.163]
Mini	Audio	shuffled	0	6000	5	0.020 [0.020, 0.020]	0.020 [0.018, 0.023]
Mini	Audio	shuffled	0	600	5	0.019 [0.015, 0.020]	0.019 [0.016, 0.024]
Mini	IMU	shuffled	0	6000	2/5	–	–
Mini	IMU	shuffled	0	600	5	0.167 [0.167, 0.167]	0.156 [0.148, 0.163]
Mini	Vision	shuffled	0	6000	2/3	–	–
Mini	Vision	shuffled	0	600	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Audio	shuffled	0	6000	5	0.020 [0.020, 0.020]	0.020 [0.017, 0.024]
Qwen 0.8B	Audio	shuffled	0	600	5	0.022 [0.017, 0.028]	0.022 [0.016, 0.030]
Qwen 0.8B	IMU	shuffled	0	6000	5	0.173 [0.165, 0.189]	0.164 [0.150, 0.179]
Qwen 0.8B	IMU	shuffled	0	600	5	0.158 [0.096, 0.219]	0.151 [0.090, 0.216]
Qwen 0.8B	Vision	shuffled	0	6000	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Vision	shuffled	0	600	3	0.123 [0.100, 0.138]	0.123 [0.100, 0.138]
LFM 230M	Audio	true	0	6000	3	0.386 [0.336, 0.442]	0.386 [0.337, 0.440]

Host	Task	Interface	λ	Step	n	Macro [95%]	Micro [95%]
LFM 230M	Audio	true	0	600	3	0.260 [0.180, 0.331]	0.260 [0.182, 0.331]
LFM 230M	IMU	true	0	6000	3	0.839 [0.761, 0.913]	0.853 [0.784, 0.920]
LFM 230M	IMU	true	0	600	3	0.834 [0.759, 0.911]	0.844 [0.773, 0.916]
LFM 230M	Audio	static	0	6000	3	0.020 [0.020, 0.020]	0.020 [0.017, 0.025]
LFM 230M	Audio	static	0	600	3	0.020 [0.020, 0.020]	0.020 [0.017, 0.024]
LFM 230M	IMU	static	0	6000	3	0.167 [0.167, 0.167]	0.154 [0.143, 0.163]
LFM 230M	IMU	static	0	600	3	0.167 [0.167, 0.167]	0.154 [0.143, 0.163]
Mini	Audio	true	0	6000	5	0.461 [0.419, 0.506]	0.461 [0.417, 0.504]
Mini	Audio	true	0	600	5	0.318 [0.279, 0.355]	0.318 [0.278, 0.357]
Mini	IMU	true	0	6000	0/5	–	–
Mini	IMU	true	0	600	5	0.759 [0.714, 0.798]	0.760 [0.711, 0.804]
Mini	Vision	true	0	6000	3	0.582 [0.526, 0.628]	0.582 [0.526, 0.628]
Mini	Vision	true	0	600	3	0.431 [0.404, 0.464]	0.431 [0.404, 0.464]
Mini	Audio	static	0	6000	5	0.020 [0.020, 0.020]	0.020 [0.018, 0.023]
Mini	Audio	static	0	600	5	0.020 [0.020, 0.020]	0.020 [0.017, 0.025]
Mini	IMU	static	0	6000	5	0.167 [0.167, 0.167]	0.156 [0.148, 0.163]
Mini	IMU	static	0	600	5	0.167 [0.167, 0.167]	0.156 [0.148, 0.163]
Mini	Vision	static	0	6000	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Mini	Vision	static	0	600	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Audio	true	0	6000	5	0.421 [0.383, 0.459]	0.421 [0.384, 0.461]
Qwen 0.8B	Audio	true	0	600	5	0.166 [0.135, 0.205]	0.166 [0.134, 0.204]
Qwen 0.8B	IMU	true	0	6000	5	0.838 [0.733, 0.927]	0.849 [0.752, 0.932]
Qwen 0.8B	IMU	true	0	600	5	0.828 [0.738, 0.918]	0.833 [0.742, 0.921]
Qwen 0.8B	Vision	true	0	6000	3	0.451 [0.353, 0.553]	0.451 [0.353, 0.553]
Qwen 0.8B	Vision	true	0	600	3	0.341 [0.305, 0.365]	0.341 [0.305, 0.365]
Qwen 0.8B	Audio	static	0	6000	5	0.020 [0.020, 0.020]	0.020 [0.017, 0.025]
Qwen 0.8B	Audio	static	0	600	5	0.020 [0.020, 0.020]	0.020 [0.017, 0.025]
Qwen 0.8B	IMU	static	0	6000	5	0.167 [0.167, 0.167]	0.153 [0.144, 0.161]
Qwen 0.8B	IMU	static	0	600	5	0.167 [0.167, 0.167]	0.160 [0.155, 0.164]
Qwen 0.8B	Vision	static	0	6000	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
Qwen 0.8B	Vision	static	0	600	3	0.100 [0.100, 0.100]	0.100 [0.100, 0.100]
LFM 230M	IMU	true	0.01	6000	3	0.768 [0.645, 0.869]	0.788 [0.676, 0.878]
LFM 230M	IMU	true	0.01	600	3	0.723 [0.644, 0.809]	0.745 [0.669, 0.826]
Mini	IMU	true	0.01	6000	0/3	–	–
Mini	IMU	true	0.01	600	3	0.802 [0.770, 0.830]	0.802 [0.765, 0.833]
Qwen 0.8B	IMU	true	0.01	6000	3	0.607 [0.553, 0.649]	0.632 [0.575, 0.682]
Qwen 0.8B	IMU	true	0.01	600	3	0.738 [0.638, 0.833]	0.760 [0.667, 0.848]
LFM 230M	IMU	true	0.05	6000	3	0.741 [0.652, 0.802]	0.762 [0.683, 0.818]
LFM 230M	IMU	true	0.05	600	3	0.759 [0.651, 0.908]	0.773 [0.670, 0.908]
Mini	IMU	true	0.05	6000	0/3	–	–
Mini	IMU	true	0.05	600	2/3	–	–
Qwen 0.8B	IMU	true	0.05	6000	3	0.762 [0.653, 0.867]	0.777 [0.676, 0.877]
Qwen 0.8B	IMU	true	0.05	600	3	0.756 [0.690, 0.821]	0.769 [0.709, 0.829]
LFM 230M	IMU	true	0.1	6000	3	0.676 [0.640, 0.729]	0.694 [0.656, 0.747]
LFM 230M	IMU	true	0.1	600	3	0.709 [0.623, 0.794]	0.730 [0.649, 0.808]
Mini	IMU	true	0.1	6000	0/3	–	–
Mini	IMU	true	0.1	600	3	0.759 [0.701, 0.804]	0.762 [0.703, 0.808]
Qwen 0.8B	IMU	true	0.1	6000	3	0.762 [0.692, 0.834]	0.763 [0.695, 0.835]
Qwen 0.8B	IMU	true	0.1	600	3	0.667 [0.484, 0.820]	0.685 [0.500, 0.831]
LFM 230M	IMU	true	0.2	6000	3	0.737 [0.658, 0.840]	0.757 [0.682, 0.846]
LFM 230M	IMU	true	0.2	600	3	0.733 [0.628, 0.820]	0.749 [0.657, 0.830]
Mini	IMU	true	0.2	6000	0/3	–	–

Host	Task	Interface	λ	Step	n	Macro [95%]	Micro [95%]
Mini	IMU	true	0.2	600	3	0.804 [0.725, 0.903]	0.803 [0.722, 0.905]
Qwen 0.8B	IMU	true	0.2	6000	3	0.838 [0.718, 0.929]	0.837 [0.712, 0.934]
Qwen 0.8B	IMU	true	0.2	600	3	0.820 [0.680, 0.923]	0.833 [0.706, 0.929]

Table 17: Feature references on the full test population. The bootstrap resamples held-out clusters while keeping the classifiers fixed; it does not resample independently trained encoders. The audio exposure described in the protocol still applies.

Task	Frozen-feature reference	Macro [95%]	Micro [95%]
Audio	Linear	0.662 [0.624, 0.700]	0.662 [0.622, 0.702]
Audio	Original head	0.652 [0.613, 0.696]	0.652 [0.610, 0.694]
IMU	Linear	0.967 [0.934, 0.989]	0.968 [0.935, 0.989]
IMU	Original head	0.966 [0.931, 0.989]	0.966 [0.931, 0.989]
Vision	Linear	0.799 [0.791, 0.807]	0.799 [0.791, 0.807]

Table 18: Audio sensitivity analysis excluding original sources shared by training and test sets. Test inputs still influenced encoder normalization.

Host	Excluded clips	All test [95%]	Source-disjoint [95%]
Mini	3	0.461 [0.419, 0.506]	0.461 [0.419, 0.506]
LFM 230M	3	0.386 [0.336, 0.442]	0.383 [0.333, 0.440]
Qwen 0.8B	3	0.421 [0.383, 0.459]	0.420 [0.382, 0.458]

Table 19: Resource measurements at batch size one on the isolated RTX PRO 6000. Parameters are millions of trainable adapter parameters. VRAM is peak allocated GiB with the host, adapter, and external encoder resident. First is median time from raw sensor input to the first token, in milliseconds; 32-token is median generation time with cached sensor factors and a fresh text cache. Both use 128 input text tokens. Frozen/static paths skip unused sensor work.

Host	Task	Interface	Params M	VRAM GiB	First ms	32-token ms
Mini	Audio	Bridge	84.35	0.49	5.2	94.3
Mini	Audio	None	0.00	0.17	1.7	48.8
Mini	Audio	Prefix	2.10	0.18	3.3	49.3
Mini	Audio	Static	0.33	0.17	2.7	82.8
Mini	IMU	Bridge	84.33	0.49	3.7	90.0
Mini	IMU	None	0.00	0.17	1.8	49.8
Mini	IMU	Prefix	1.05	0.18	2.0	49.2
Mini	IMU	Static	0.33	0.17	2.7	82.3
Mini	Vision	Bridge	84.35	0.51	5.2	89.9
Mini	Vision	None	0.00	0.19	1.7	47.8
Mini	Vision	Prefix	2.10	0.20	3.5	50.5
Mini	Vision	Static	0.33	0.19	2.8	83.1
LFM 230M	Audio	Bridge	199.09	1.67	8.0	160.3
LFM 230M	Audio	None	0.00	0.93	4.2	113.6
LFM 230M	Audio	Prefix	4.20	0.95	6.3	116.0

Host	Task	Interface	Params M	VRAM GiB	First ms	32-token ms
LFM 230M	Audio	Static	0.77	0.93	5.6	156.6
LFM 230M	IMU	Bridge	199.07	1.67	6.5	161.0
LFM 230M	IMU	None	0.00	0.93	4.2	113.4
LFM 230M	IMU	Prefix	2.11	0.94	4.7	114.0
LFM 230M	IMU	Static	0.77	0.93	5.7	158.6
LFM 230M	Vision	Bridge	199.09	1.69	8.1	160.8
LFM 230M	Vision	None	0.00	0.94	4.2	115.2
LFM 230M	Vision	Prefix	4.20	0.96	6.3	114.1
LFM 230M	Vision	Static	0.77	0.94	5.5	152.2
Qwen 0.8B	Audio	Bridge	410.67	4.52	52.6	431.0
Qwen 0.8B	Audio	None	0.00	2.99	47.4	373.3
Qwen 0.8B	Audio	Prefix	4.20	3.01	52.5	361.7
Qwen 0.8B	Audio	Static	1.60	2.99	49.3	425.0
Qwen 0.8B	IMU	Bridge	410.66	4.52	50.6	440.1
Qwen 0.8B	IMU	None	0.00	2.99	46.8	359.8
Qwen 0.8B	IMU	Prefix	2.11	3.00	50.8	376.8
Qwen 0.8B	IMU	Static	1.60	2.99	49.4	423.0
Qwen 0.8B	Vision	Bridge	410.67	4.54	52.1	440.0
Qwen 0.8B	Vision	None	0.00	3.00	46.8	362.1
Qwen 0.8B	Vision	Prefix	4.20	3.02	51.6	362.9
Qwen 0.8B	Vision	Static	1.60	3.00	48.6	417.9